Matching Diverse Types

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Abstract

Lehrer and Pauzner (1999) show that the set of subgame perfect equilibrium values expands as we increase the discount rate of the more patient player or reduce the discount rate of the less patient player, provided interactions are sufficiently frequent. We show that the payoff set for any matched pair also expands as we increase the type gap in a learning model when agent's types encode their horizontally differentiated information sources. In both case, the Pareto frontier for matched pairs is increasing in type *diversity*.

We explore the implications of such a preference for type diversity in general imperfectly transferable utility matching models. Firstly, equilibrium matchings must be *negative quadrant supported (NQS)*; namely all players below (above) a given cutoff type must be matched to players with a higher (lower) type. Furthermore, equilibrium values must be quasiconvex in types and minimized at the cutoff type. In the special case of non-transferable utility, preferences for diversity imply negative assortative matching, but in general NQS does not imply negative assortative matching (NAM). In fact, positive sorting can obtain within each matching quadrant (a pattern we call NS-PAM). However, we show that negative sorting on discount rates obtains for all frequently repeated games with finite pure strategies in the stage game, and provide sufficient conditions for NAM and NS-PAM in our matching information sources application.

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1 Introduction

Many economic interactions are more valuable when agents have different characteristics. Research collaborations benefit from diverse expertise, repeated cooperation may be easier to sustain between agents with different preferences, and organizations often gain from combining heterogeneous information sources. In these examples, characteristics are horizontally differentiated and the value of a match increases in the diversity of characteristics within a match. We characterize equilibrium matching patterns and payoffs in such environments.

We explore bilateral two-sided matching with imperfectly transferable utility (ITU) as in Legros and Newman (2007a), where the payoffs within a match are described by a utility possibility frontier. Assuming a linear frontier induces perfectly transferable utility (TU), while non-transferable utility (NTU) is the limit case where the frontier collapses to a single point. We label the two sides rows and columns, anticipating our main application of matching to play repeated games. We model preferences for diversity, as a monotonicity property of the utility possibility frontier: holding one type fixed, moving the other type farther away expands the frontier. Since this is a first derivative property, it is orthogonal to the standard complementarity between types for TU models in Becker (1973), and also the generalized monotone differences condition for ITU models in Legros and Newman (2007a).

Preferences for diversity have several equilibrium implications. First, matchings must be *negative quadrant supported* (NQS): all types below (above) a threshold type are matched to agents with a higher (lower) type (Proposition 1). Thus, every matched pair must lie in either a northwest quadrant everywhere above the 45 degree line or southeast quadrant everywhere below the 45 degree line. Second, if the masses of row and column types are unequal, then unmatched agents on the long side form a single interval that is common to all equilibria (Proposition 2). Third, values are quasiconvex: minimized at a threshold type, and rising from the threshold type to the highest and lowest types (Proposition 3). Intuitively, extreme types can form more diverse matches and generate higher frontier payoffs, and thus can command higher equilibrium payoffs. Fourth, equilibrium values have opposite monotonicity within each matching quadrant. That is, if equilibrium values are increasing in row types within a matching quadrant, they must be decreasing in column types in this same quadrant.

For NTU models we establish a stark sorting result (Proposition 4): the unique equilibrium matching involves perfect negative sorting on types (NAM). Intuitively, if all participants strictly prefer partners with types further from their own, the most extreme types on the two sides of the market are mutually ideal partners, and so must match. Once these types have been matched, the most extreme types of those remaining are mutually ideal partners; and

so on, implying that NAM is the only stable outcome.

This sharp sorting result does not hold under ITU without further assumptions, since equilibrium sorting depends not only on how marginal productivity responds to partner types, but also on how the degree of transferability varies with types. Legros and Newman (2007a) show that the generalized decreasing differences (GDD) condition, which captures both forces, is sufficient for an economy to *exhibit NAM*: a negatively assortative equilibrium exists, and every equilibrium is payoff equivalent to one where the matching is NAM. The opposite condition, generalized increasing differences (GID) implies that the economy exhibits perfectly positive sorting (PAM). Under mild regularity conditions, GDD holds when a row agents willingness to pay for an increase in their partner's type is decreasing in their own (row) type. The reverse monotonicity condition yields generalized increasing differences (GID).

While preferences for diversity do not by themselves pin down sorting, they impose a subtle restriction on how GID may apply. In particular, preferences for diversity are incompatible with GID whenever the 45-degree line is in the interior of the matching support. Thus, global PAM is generally ruled out. Nevertheless, it is still possible that the GID is separately satisfied both above and below the 45 degree line, in which case the matching must be NQS with positive sorting within each of the negative quadrants (NS-PAM) as in the left panel below. In contrast, if GDD is satisfied for types above and below the 45 degree line, negative sorting is implied in each matched quadrant thus yielding NAM.

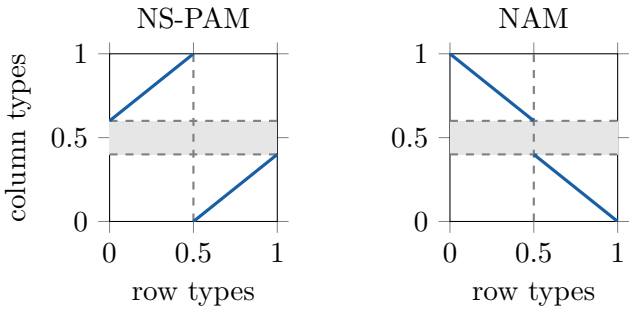


Figure 1: The extreme NQS matchings (blue lines) with middle types (gray) unmatched.

One of our contributions is to identify a broad environment in which preferences for diversity are satisfied: matching to play repeated games. Assume that types are discount factors, and that rows and columns form pairwise matches to play an infinitely repeated stage game with finite pure strategies. The utility possibility frontier is then the frontier of the set of subgame perfect equilibrium values. Lehrer and Pauzner (1999) solve for the SPE payoff frontier for continuously repeated games played between agents with different discount

factors. This frontier expands as we increase the discount factor of the more patient player or reduce the discount factor of the less patient player, thus satisfying preferences for diversity. Intuitively, differences in time preferences create opportunities for intertemporal trade, since an impatient player values early payoffs relatively more, while a patient player values later payoffs relatively more.[1] We further show that the payoff frontier satisfies GDD restricted to type pairs above and below the 45 degree line; and thus, since the matching support cannot include the 45 degree line, the economy must exhibit NAM (Proposition 5). When outside options are sufficiently high, NAM is the unique equilibrium. All of these results obtain in discretely repeated games with sufficiently frequent interactions.

We explore a second application with more nuanced sorting results: matching information sources. Our application is adapted from Chade and Eeckhout (2018), who study TU matching where match surplus is an expected payoff generated by a common action given an uncertain state of the world. Prior to taking an action, each agent receives an informative signal of the state with precision corresponding to the agent's type and a constant correlation across all possible matched pairs. In their framework, types are vertically differentiated, since higher signal precision is always better. We consider a complimentary information aggregation model with horizontally differentiated types: All agents share a common signal precision, but signals become less correlated as the distances between matched types increases. Intuitively, types parameterize information sources, and pairs with more diverse information sources aggregate less redundant information and therefore generate greater total surplus. We also generalize to ITU by separately defining the payoff to each agent as a function of the common action and assuming heterogeneity in preferences over state contingent actions. Unlike in the matching to play repeated games application, NAM is not guaranteed when matching diverse sources of information. In the TU version of the model, we derive a simple inequality that is sufficient for NAM, satisfied whenever the signal correlation is concave or not too convex in the distance between types. But when the signal correlation is sufficiently convex in type diversity (e.g., log-convex), Legros and Newman's GID condition holds separately on each side of the diagonal; and the matching is NS-PAM. For the ITU model we derive an analogous sufficient condition for NAM that is satisfied by a larger set of signal correlation functions. The model nests NTU and TU as limiting cases: as the preference bias vanishes, the payoff set collapses to a point and NAM obtains; as the preference bias becomes large, the TU conditions for NAM and NS-PAM apply in the limit (Proposition 6).

[1]Such gains are supported by Agranov, Kim, and Yariv (2024), whose experimental data show that even small discount-factor differences can increase repeated game payoffs, while larger differences generate substantial inter-temporal trade, with the impatient player front-loaded and the patient player compensated later.

Additional Related Literature. A strand of the literature considers matching when payoffs *decrease* in diversity. Klumpp (2009) assumes that surplus decreases in the difference in types. Under NTU, recursively matching the closest available pairs produces stable assignments, but this recursive procedure does not generally lead to PAM. This contrasts with our result that preferences for diversity imply NAM in NTU models. The intuition for the distinction is that regardless of the distributions over types, iterative elimination of the pair furthest apart preserves negative sorting among all matched pairs, whereas the converse does not hold for PAM. Klumpp also shows that every TU-stable assignment is also stable under NTU, provided that surplus is sufficiently convex in the distance between types.

Clark (2020) also studies TU matching when match surplus is a decreasing function of the type difference. Positive sorting obtains when output is concave in the gap or the two distributions coincide. As in our model, equilibrium values rise on one side exactly where they fall on the other. In our model, this result is driven by the fact that payoff frontiers must move in opposite directions in an asymmetric match, while in Clark this result is driven by local scarcity, embodied by the difference between the densities on each side of the market. Boerma et al. (2023) derives a TU model in which match surplus is decreasing and convex in the type difference; and thus, neither supermodular nor submodular. Consistent with Clark (2020), the optimal assignment need not be monotonically sorted but exhibits composite sorting: distinct worker types can occupy the same occupation, and a worker type can participate in both positively and negatively sorted matches.

Our results also complement existing characterizations of sorting. First, our NS-PAM pattern, with positive sorting within each of the two negative quadrants, is related to median matching in Legros and Newman (2002). In their symmetric, one-sided TU economy, median matching pairs each type in the upper half of the distribution with the type of the same within-half quantile in the lower half. Translated to a balanced two-sided market with identical type distributions, this assignment is a special case of our NS-PAM. Second, our result connects preferences for diversity to established conditions for assortative matching under NTU. Becker (1973) notes that PAM obtains when every agent strictly prefers a higher-type partner under NTU; similarly, NAM obtains when one side prefers higher-type partners while the other prefers lower-type partners. Legros and Newman (2010) show that globally monotone partner rankings are not necessary for assortative matching. Their *co-ranking* condition provides a necessary and sufficient condition for PAM across type distributions: in every two-by-two submarket, either the higher-type pair or the lower-type pair must mutually prefer one another. Reversing the type ordering on one side yields the analogous characterization of

NAM: either of the two oppositely ranked pairs must mutually prefer one another. Our preferences-for-diversity condition implies this reverse co-ranking property. Thus, our NTU result identifies an economically interpretable sufficient condition for NAM.

Demange and Gale (1985) consider bilateral ITU matching. They show that if an agent is unmatched for some stable payoff, then for any other stable payoff, there exists a stable matching that is compatible with it and leaves the agent unmatched. Their result requires a full-range assumption on payoffs, while we exploit the matching structure and monotonicity properties of value functions implied by preferences for diversity. Moreover, we show that the unmatched set is the same across *all* equilibria.

Our information application also relates to Kambhampati et al. (2024), who, like us, fix precision and vary correlation across pairs. Their model also allows agents to choose how many signals to acquire at linear cost, but they assume correlation takes one of two values, high or low. They show that maximizing the number of low correlation pairs maximizes total surplus; and that moral hazard in signal acquisition leads to too many high correlation pairs in equilibrium. However, their model cannot speak to sorting across *types of agents*, as there is no natural ordering of types.

Matching to share risk can induce preferences for diversity, but need not do so. In Legros and Newman (2007a) individuals with different risk tolerance match to share exogenous risk in a two state world. They show that the strict GDD condition is globally satisfied and so any stable match must be negative assortative. Intuitively, more risk-averse agents have a greater willingness to pay for insurance, while less risk-averse partners are better placed to provide it. They provide a closed-form frontier example that satisfies GDD, but not a preference for diversity. Our information application allows the converse: preferences for diversity without GDD. Chiappori and Reny (2016) extend the Legros and Newman (2007a) negative sorting result to a risk-sharing model with an arbitrary number of states. Neither paper characterizes values or the set of unmatched types.

Moscariello (2026) considers gerrymandering in two party systems with primaries. The paper proves that the optimal gerrymandering problem can be reinterpreted as an optimal TU matching of voters with ideal points above the median to voters with ideal points below the median, and that the resulting match surplus exhibits a preference for diversity. NS-PAM (NAM) obtains when the cdf over preference shocks is concave (convex).

Section 2 gives the model, Section 3 characterizes matching and values with a preference for diversity, Section 4 gives sufficient conditions for NAM and NS-PAM, Section 5 explores matching to play repeated games, and Section 6 considers matching information sources.

2 The Model

Anticipating our main application, we describe the general model using the language of matching to play games. The economy is divided ex-ante into a population of row players and a population of column players.[2] The ratio of column players to row players is $K \geq 1$. Within each population, players are identical except for their scalar types (discount factors). The cumulative distribution function over discount factors is H_R for row players with representative type β and H_C among column players with representative type δ. H_R and H_C have full support on $[\underline{\beta}, \bar{\beta}]$ and $[\underline{\delta}, \bar{\delta}]$, respectively, both strict subsets of $[0, 1]$. We assume each distribution is absolutely continuous.[3]

An *outcome* describes which column players remain unmatched and the pairing among the types that do match. Assume no unmatched row players (we argue below that this is WLOG). Since column players weakly outnumber row players, fraction $(K-1)/K$ of column players must be single. Denoting the cdf over discount factors for unmatched column players as S_C, the cdf over matched column players $\hat{H}_C$ obeys:

$$\frac{K-1}{K} S_C + \frac{1}{K} \hat{H}_C = H_C \quad \Rightarrow \quad \hat{H}_C = K H_C - (K-1) S_C \tag{1}$$

The *matching* of players is a bivariate cdf M on $[\underline{\beta}, \bar{\beta}] \times [\underline{\delta}, \bar{\delta}]$ with marginals H_R and $\hat{H}_C$. The matching is *pure* if it is described by a function μ from row types onto column types. We say that the pair (β, δ) is *matched* if (β, δ) is in the support of M. If M is a pure matching described by μ, then (β, δ) are matched if and only if $\delta = \mu(\beta)$. *Negative assortative matching* (NAM) is the pure matching in which complementary quantiles match, while *positive assortative matching* (PAM) is a pure matching in which like quantiles match. Thus, fixing the cdf over matched column types $\hat{H}_C$, the NAM and PAM matching are the increasing functions μ_N and μ_P that uniquely solve:

$$H_R(\beta) = 1 - \hat{H}_C(\mu_N(\beta)) \quad \text{and} \quad H_R(\beta) = \hat{H}_C(\mu_P(\beta)) \tag{2}$$

In particular, given uniform distributions over types with the same domain and $K = 1$, the NAM matching function is $\mu_N(\beta) = 1 - \beta$ and $\mu_P(\beta) = \beta$. Note that when $K > 1$, these matching functions will depend on the distribution of single column types S_C.

Let $V(\beta, \delta) \subset \mathbb{R}^2$ be the non-empty, compact set of possible (row,column) payoffs when

[2] We focus on this two-sided markets case, but our results also apply when all players are drawn from a common population, provided that the set of payoffs is symmetric across the row and column roles.

[3] Our results also hold in *discrete type* matching models, i.e. where there are a finite number of distinct types on each side of the market, except for value uniqueness in Corollary 3.

a row player of type β is matched to a column player of type δ. We assume $V(\beta,\delta)$ has a continuous non-degenerate Pareto frontier, i.e. that the lowest and higher frontier payoffs for row obey $\underline{u}(\beta,\delta) < \bar{u}(\beta,\delta)$; and thus, the highest and lowest frontier payoffs for column obey $\underline{v}(\beta,\delta) < \bar{v}(\beta,\delta)$). Let $\phi(\beta,\delta,w)$ denote the frontier row payoff, i.e. the highest row payoff in $V(\beta,\delta)$ given that the column payoff is $w \in [\underline{v}(\beta,\delta), \bar{v}(\beta,\delta)]$. ϕ is strictly decreasing in w, so the *partial inverse* ψ is well defined by $\phi(\beta,\delta,\psi(\delta,\beta,\nu)) \equiv \nu$ for all $\nu \in [\underline{u}(\beta,\delta), \bar{u}(\beta,\delta)]$. We assume the Pareto frontier is continuously differentiable, except perhaps at $\beta = \delta$. Single (row, column) players earn outside options (u_0, v_0) with $u_0 < \phi(\beta,\delta,v_0)$ for all $(\beta,\delta) \in [\underline{\beta},\bar{\beta}] \times [\underline{\delta},\bar{\delta}]$ (outside options are feasible payoffs for any matched pair), and $(u_0,v_0) \geq (\underline{u},\underline{v})$.[4]

An *allocation* is a tuple (S_C, M, u, v), where M is a matching given S_C, and $u : [\underline{\beta},\bar{\beta}] \mapsto \mathbb{R}$ and $v : [\underline{\delta},\bar{\delta}] \mapsto \mathbb{R}$ are the payoff functions for the row and column players as a function of their types. An allocation is *feasible* if $(u(\beta), v(\delta)) \in V(\beta,\delta)$ for all matched pairs (β,δ), and $v(\delta) = v_0$ for all $\delta \in \text{supp}(S_C)$. An allocation is *blocked* by pair (β,δ) if $\exists (u',v') \in V(\beta,\delta)$ with $u' > u(\beta)$ and $v' > v(\delta)$. An allocation is *stable* if it is not blocked by any $(\beta,\delta) \in [\underline{\beta},\bar{\beta}] \times [\underline{\delta},\bar{\delta}]$ and $(u,v) \geq (u_0,v_0)$. An allocation (S_C, M, u, v) is an *equilibrium* if it is feasible and stable.

There are two immediate implication of feasibility and stability. Firstly, no row players can be unmatched in equilibrium, since such an equilibrium would be blocked by any unmatched row and column player, since (u_0,v_0) in the interior of $V(\beta,\delta)$. The second immediate implication is that the payoffs for any matched pair (β,δ) must be on the frontier of $V(\beta,\delta)$ (e.g., if the payoff $(u(\beta),v(\delta))$ is strictly below the frontier for any matched pair (β,δ), then that same matched pair would block the allocation). Thus, in any equilibrium allocation:

$$u(\beta) = \max_{\delta \in [\underline{\delta},\bar{\delta}]} \phi(\beta,\delta,v(\delta)) \;\; \forall \beta \in [\underline{\beta},\bar{\beta}] \qquad v(\delta) = \max_{\beta \in [\underline{\beta},\bar{\beta}]} \psi(\delta,\beta,u(\beta)) \;\; \forall \delta \in [\underline{\delta},\bar{\delta}] \tag{3}$$

and

$$(\beta',\delta') \in \text{supp}(M) \quad \Rightarrow \quad \delta' \in \arg\max_{\delta \in [\underline{\delta},\bar{\delta}]} \phi(\beta',\delta,v(\delta)) \text{ and } \beta' \in \arg\max_{\beta \in [\underline{\beta},\bar{\beta}]} \psi(\delta',\beta,u(\beta)) \tag{4}$$

This imperfectly transferable utility (ITU) model subsumes perfectly transferable utility (TU) as a special case. Specifically, if we assume that matched pairs produce a perfectly divisible output $g(\beta,\delta)$, then this defines a Pareto frontier $\phi(\beta,\delta,w) = g(\beta,\delta) - w$.

[4]The lower bound on v_0 is necessary for existence when $K > 1$. For if $\underline{v} > v_0$, then matched column gets a payoff strictly above v_0, but then unmatched column players can form blocking pair with a matched row player by offering the row player at least his current payoff while accepting a payoff strictly between v_0 and that of the currently matched column player. The lower bound on u_0 is only used to secure uniqueness of the set of unmatched columns in Proposition 2.

3 Matching and Values with a Preference for Diversity

We now explore equilibrium matching patterns and values when matching unlike types is beneficial. Specifically, we henceforth assume *preferences for diversity*, i.e.

$$\beta > \delta \;\Rightarrow\; \phi_1(\beta,\delta,w) > 0 > \phi_2(\beta,\delta,w) \quad \text{and} \quad \beta < \delta \;\Rightarrow\; \phi_1(\beta,\delta,w) < 0 < \phi_2(\beta,\delta,w) \tag{5}$$

Note that (5) do not directly rank partners on opposite sides of an agent's type. For example, if $\delta < \beta < \delta'$, (5) alone does not imply that β prefers the column type further away from her.

A. Matching Implications. The matching is *Negative Quadrant Supported (NQS)* when the set of unmatched column types is an interval (δ_L^*, δ_H^*), all $\delta \leq \delta_L^*$ match up to some $\beta \geq \beta^*$, and all $\delta \geq \delta_H^*$ match down to some $\beta \leq \beta^*$. As a result, the matching support is a subset of the northwest quadrant $[\underline{\beta}, \beta^*] \times [\delta_H^*, \bar{\delta}]$ and the southeast quadrant $[\beta^*, \bar{\beta}] \times [\underline{\delta}, \delta_L^*]$.

Figure 2 illustrates possible supports for β uniform on $[0.3, 0.9]$ and δ uniform on $[0.1, 0.7]$. In the left graph, there is an equal mass of rows and columns and $\beta^* = \delta^* = 0.5$. On the right, columns outnumber rows ($K > 1$), and middle column types are unmatched. In both panels, since all $\beta < \beta^*$ match up and all $\beta > \beta^*$ match down, we cannot have any matched pairs in the blue region. Similarly, since low δ match up and high δ match down, no match can form in the green region. The gray band denotes unmatched column types. Altogether, matched pairs can only lie in the unshaded NW and SE regions.

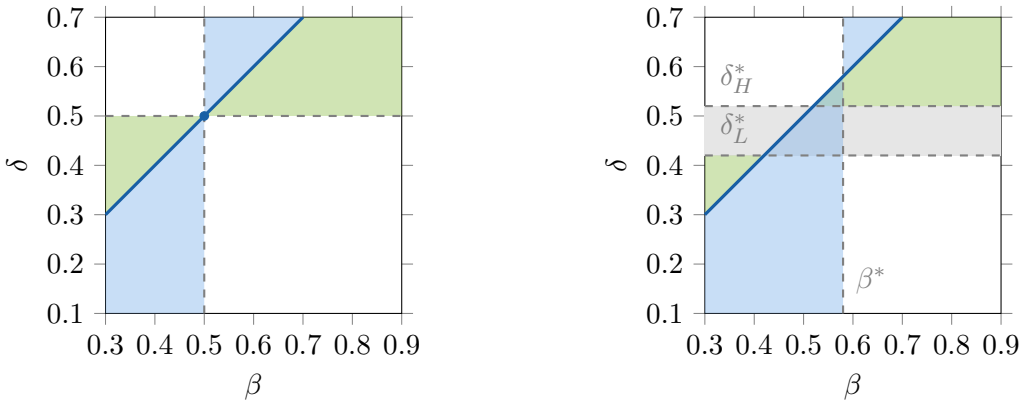


Figure 2: Examples of Negative Quadrant Supports for Matchings.

Proposition 1 *Equilibrium matchings must be negative quadrant supported.*

Proof: Throughout we fix an equilibrium with column payoff function $v(\cdot)$.

STEP 1: THE UNMATCHED SET IS AN INTERVAL (δ_L^*, δ_H^*).
Toward a contradiction assume that two column types $\delta_L < \delta_H$ are unmatched while some intermediate type $\delta' \in (\delta_L, \delta_H)$ is matched with row type β'. We show that this is impossible for $\beta' \geq \delta' > \delta_L$, the case of $\beta' < \delta' < \delta_H$ is symmetric. Indeed, use the fact that unmatched column types receive their outside option v_0, followed by $\phi_3 < 0$ along with $v \geq v_0$, and then payoffs increasing in diversity with $\beta' \geq \delta' > \delta_L$ to deduce:

$$\phi(\beta', \delta_L, v(\delta_L)) = \phi(\beta', \delta_L, v_0) \geq \phi(\beta', \delta_L, v(\delta')) > \phi(\beta', \delta', v(\delta'))$$

But $\phi(\beta', \delta_L, v(\delta_L)) > \phi(\beta', \delta', v(\delta'))$ contradicts (β', δ') matched and maximization (4).

STEP 2: THERE EXISTS β^* S.T. $\beta < \beta^*$ MATCHES UP AND $\beta > \beta^*$ MATCHES DOWN.
Toward a contradiction, assume $\beta < \beta'$ such that β matches down (to δ) and β' matches up (to δ'). Since payoffs increase in diversity and $\delta \leq \beta < \beta' \leq \delta'$:

$$\phi(\beta', \delta', v(\delta')) < \phi(\beta, \delta', v(\delta')) \quad \text{and} \quad \phi(\beta, \delta, v(\delta)) < \phi(\beta', \delta, v(\delta)) \tag{6}$$

Now use the left inequality in (6), followed by (β, δ) matched and maximization (4), and then the right inequality in (6) to find:

$$\phi(\beta', \delta', v(\delta')) < \phi(\beta, \delta', v(\delta')) \leq \phi(\beta, \delta, v(\delta)) < \phi(\beta', \delta, v(\delta))$$

But then δ' cannot be matched to β' by (4).

STEP 3: ALL $\delta < \delta_L^*$ MATCH UP AND ALL $\delta > \delta_H^*$ MATCH DOWN.
Parallel logic to that used in Step 2, establishes the existence of a cutoff δ^* such that, *conditional on matching*, all types below δ^* match up and types above δ^* match down. This establishes the result for $K = 1$ (with $\delta^* = \delta_L^* = \delta_H^*$). We now show that all $\delta < \delta_L^*$ match up when $\delta_L^* < \delta_H^*$. Parallel logic establishes that all $\delta > \delta_H^*$ match down when $\delta_L^* < \delta_H^*$.

Toward a contradiction, assume some $\delta < \delta_L^*$ matches down to β. Then use the fact that unmatched column types receive their outside option v_0, followed by $\phi_3 < 0$ along with $v \geq v_0$, and then payoffs increasing in diversity with $\delta_L^* > \delta \geq \beta$ to deduce:

$$\phi(\beta, \delta_L^*, v(\delta_L^*)) = \phi(\beta, \delta_L^*, v_0) \geq \phi(\beta, \delta_L^*, v(\delta)) > \phi(\beta, \delta, v(\delta))$$

But this contradicts (β, δ) matched and maximization (4). □

For any candidate cutoff $\beta^* = \beta$ in a NQS equilibrium, the boundaries of the unmatched

column interval, $\delta_L(\beta) < \delta_H(\beta)$ must obey the following *market clearing conditions*:

$$H_R(\beta) = K\left(1 - H_C(\delta_H(\beta))\right) \quad \text{and} \quad 1 - H_R(\beta) = K H_C(\delta_L(\beta)) \tag{7}$$

Since H_R and H_C admit densities, these functions are uniquely solvable, and $\delta_L < \delta_H$ are strictly decreasing and differentiable with endpoints:

$$\delta_H(\underline{\beta}) = \bar{\delta} > H_C^{-1}\left(\frac{1}{K}\right) = \delta_L(\underline{\beta}) \quad \text{and} \quad \delta_H(\bar{\beta}) = H_C^{-1}\left(\frac{K-1}{K}\right) > \underline{\delta} = \delta_L(\bar{\beta}) \tag{8}$$

Now define the payoff to any β from matching with the threshold types $\delta_H(\beta)$ and $\delta_L(\delta)$:

$$\phi_H(\beta) \equiv \phi(\beta, \delta_H(\beta), v_0) \quad \text{and} \quad \phi_L(\beta) \equiv \phi(\beta, \delta_L(\beta), v_0)$$

Each of these functions inherits differentiability and continuity from ϕ, δ_L, and δ_H.

Lemma 1 *The difference $\phi_H(\beta) - \phi_L(\beta)$ is strictly down-crossing in β when $K > 1$.*

Proof: We prove the result by showing that $\phi_H(\beta) = \phi_L(\beta)$ implies $\phi'_H(\beta) < \phi'_L(\beta)$.

Suppose $\phi(\beta, \delta_H(\beta), v_0) = \phi(\beta, \delta_L(\beta), v_0)$ with $\delta_H(\beta) > \delta_L(\beta)$. Only one configuration is consistent with this equality: we cannot have both $\delta_L, \delta_H \geq \beta$ (since then $\phi_2 > 0$ forces $\phi(\beta, \delta_H, v_0) > \phi(\beta, \delta_L, v_0)$, a contradiction) nor both $\leq \beta$ (since $\phi_2 < 0$ gives the reverse inequality). Hence it must be the case $\delta_H(\beta) > \beta > \delta_L(\beta)$. At δ_H: $\phi_1 < 0 < \phi_2$, so $\phi'_H(\beta) < 0$. At δ_L: $\phi_1 > 0 > \phi_2$, so $\phi'_L(\beta) > 0$. Taking together, we have $\phi'_H(\beta) < \phi'_L(\beta)$. □

Proposition 2 *If $K > 1$, every equilibrium has the same unmatched interval, which obeys*

1. *If $\phi_H(\underline{\beta}) \leq \phi_L(\underline{\beta})$, then high types are unmatched: $\delta_L^* = H_C^{-1}(1/K)$, $\delta_H^* = \bar{\delta}$, $\beta^* = \underline{\beta}$.*
2. *If $\phi_H(\bar{\beta}) \geq \phi_L(\bar{\beta})$, then low types are unmatched: $\delta_L^* = \underline{\delta}$, $\delta_H^* = H_C^{-1}(\frac{K-1}{K})$, $\beta^* = \bar{\beta}$.*

Proof: Step 1: We prove the uniqueness of the unmatched interval in Appendix A.1 .
Step 2: Unmatched column characterization. The two cases are mutually exclusive by Lemma 1. We prove the first case. The second follows symmetric reasoning.

Toward a contradiction assume $\delta_H^* < \bar{\delta}$; and thus, $\delta_H^* = \delta_H(\beta^*) < \bar{\delta}$ implies $\beta^* > \underline{\beta}$. Now, let β' be the equilibrium match for δ_H^*. Since δ_H^* matches down and δ_H is decreasing with $\beta' \leq \beta^*$, we have $\beta' \leq \delta_H^* \leq \delta_H(\beta')$ with the second inequality strict when $\beta' < \beta^*$. Furthermore, we must have $\beta' > \delta_L(\beta')$, since otherwise we would have $\beta' \leq \delta_L(\beta') < \delta_H(\beta')$, but then by payoffs increasing in diversity, β' gets a higher payoff matched to $\delta_H(\beta')$ than

matched to $\delta_L(\beta')$, contradicting $\phi_H(\beta') \leq \phi_L(\beta')$ by Lemma 1 and $\beta' \geq \underline{\beta}$. Then, since δ_L is decreasing and $\beta^* \geq \beta'$, we have $\delta_L^* \leq \delta_L(\beta') < \beta'$. Now use this inequality and payoffs increasing in diversity, followed by $\phi_H(\underline{\beta}) \leq \phi_L(\underline{\beta})$ along with Lemma 1, and then $\beta' \leq \delta_H^* \leq \delta_H(\beta')$ and payoffs increasing in diversity (a second time) to deduce:

$$\phi(\beta', \delta_L^*, v_0) \geq \phi(\beta', \delta_L(\beta'), v_0) \geq \phi(\beta', \delta_H(\beta'), v_0) \geq \phi(\beta', \delta_H^*, v_0) \tag{9}$$

At least one of these inequalities must be strict: If $\beta' = \beta^*$, then the middle inequality is strict since $\phi_L(\beta) > \phi_H(\beta)$ for all $\beta > \underline{\beta}$ and $\beta^* > \underline{\beta}$, while the first and last are strict when $\beta' < \beta^*$, since δ_L and δ_H are strictly decreasing in β. In either case, we have $\phi(\beta', \delta_L^*, v_0) > \phi(\beta', \delta_H^*, v_0)$ contradicting β' matched to δ_H^* by (4). Thus, $\delta_H^* = \bar{\delta}$ and all row types must match down to columns in $[\underline{\delta}, \delta_L^*]$, i.e. $\beta^* = \underline{\beta}$ and (by (8)) $\delta_L^* = H_C^{-1}(1/K)$. □

Cases 1 and 2 are not exhaustive, as they omit the third possibility: $\phi_H(\underline{\beta}) > \phi_L(\underline{\beta})$ and $\phi_H(\bar{\beta}) < \phi_L(\bar{\beta})$. This third case is necessary for middle types to remain unmatched; and also sufficient when NAM obtains as we will show in Section 4 (Lemma 4).

B. Value Implications. Preferences for diversity restrict how equilibrium values vary across types. Intuitively, intermediate types are less attractive because they generate smaller type diversity with potential partners. The next result formalizes this intuition.

Proposition 3 *If (S_c, M, u, v) is an equilibrium then:*
(i) $\beta' > \beta \geq \beta^* \Rightarrow u(\beta') > u(\beta)$ *and* $\delta' > \delta \geq \delta_H^* \Rightarrow v(\delta') > v(\delta)$*; and*
(ii) $\beta < \beta' \leq \beta^* \Rightarrow u(\beta') < u(\beta)$ *and* $\delta < \delta' \leq \delta_L^* \Rightarrow v(\delta') < v(\delta)$.

PROOF: We prove the result for $\beta' > \beta \geq \beta^*$. The remaining cases follow symmetric steps.

Let (S_c, M, u, v) be an equilibrium allocation. Take any $\beta' > \beta \geq \beta^*$. Since the equilibrium matching is negative quadrant supported, any $\hat{\delta}$ that is matched with β must obey $\beta \geq \hat{\delta}$. Applying maximization followed by payoffs increasing in diversity with $\beta' > \beta \geq \hat{\delta}$ for any such match partner $\hat{\delta}$ for β:

$$u(\beta') = \max_{\delta} \phi(\beta', \delta, v(\delta)) \geq \phi(\beta', \hat{\delta}, v(\hat{\delta})) > \phi(\beta, \hat{\delta}, v(\hat{\delta})) = u(\beta)$$

where the final equality follows from (3) and (4) and $(\beta, \hat{\delta}) \in \text{supp}(M)$. □

Notice that Proposition 3 implies that value functions must have the opposite monotonicity within each matching quadrant. For example, in quadrant $[\underline{\beta}, \beta^*] \times [\delta_H^*, \bar{\delta}]$, u is falling in β and v is rising in δ. Since monotone functions are almost everywhere differentiable and pay-

offs are separately monotone below and above the cutoff types, equilibrium value functions must be a.e. differentiable with derivatives given by the Envelope Theorem applied to (3).

Corollary 1 *Equilibrium values are almost everywhere differentiable with derivatives:*

$$(\beta, \delta) \in supp(M) \quad \Rightarrow \quad u'(\beta) = \phi_1(\beta, \delta, v(\delta)) \text{ and } v'(\delta) = \psi_1(\delta, \beta, u(\beta)) \tag{10}$$

C. A TU Example. Although our general model allows for imperfect transferability, we illustrate our results with a perfectly transferable utility (TU) example that admits a simple closed form solution. Specifically, assume the Pareto frontier is $\phi(\beta, \delta, w) = |\beta - \delta|^\alpha - w$ with $\alpha > 0$. Since ϕ is strictly increasing in the type gap, Proposition 1 implies that every equilibrium matching is negative quadrant supported. Moreover, $\phi_{12}(\beta, \delta) = \alpha(1-\alpha)|\beta - \delta|^{\alpha-2}$. Hence, for $\beta \neq \delta$, the surplus is locally submodular when $\alpha > 1$ and locally supermodular when $\alpha < 1$. Thus, by standard reasoning, types must be negatively (positively) sorted within each negative quadrant when $\alpha > 1$ ($\alpha < 1$). Easily, negative quadrant supported plus negative sorting within each quadrant implies global NAM. But, negative quadrant supported with positive sorting in each negative quadrant is not global PAM. We refer to a matching that is negative quadrant supported while exhibiting positive sorting within each matched quadrant as *negative quadrant supported PAM (NS-PAM).*

Now assume row types are uniformly distributed on $[0, 1]$ with total mass one, and column types are uniformly distributed on $[0, 1]$ with total mass $K = 2$. Hence one-half of the column types remain unmatched. For any given lower cutoff δ_L of the unmatched columns, market clearing determines the other two cutoffs:

$$\delta_H = \delta_L + \frac{K-1}{K} = \delta_L + \frac{1}{2}, \qquad \beta^* = 1 - K\delta_L = 1 - 2\delta_L, \qquad \delta_L \in \left[0, \frac{1}{2}\right].$$

The endpoints $\delta_L = 0$ and $\delta_L = 1/2$ correspond, respectively, to the corner cases with low and high column types being unmatched. An interior δ_L gives a middle unmatched interval.

When $\alpha = 2$, the surplus is submodular, implying NAM. Thus the matching function is

$$\mu_{\alpha=2}(\beta) = \begin{cases} 1 - \dfrac{\beta}{2}, & 0 \leq \beta \leq 1 - 2\delta_L, \\ \dfrac{1-\beta}{2}, & 1 - 2\delta_L \leq \beta \leq 1. \end{cases}$$

Substituting the induced matching function into the total surplus and integrating over the matched pairs yields the aggregate surplus $(1 + 6\delta_L - 12\delta_L^2)/4$, which is uniquely maximized at $\delta_L^* = \frac{1}{4}$, implying $\delta_H^* = \frac{3}{4}$ and $\beta^* = \frac{1}{2}$. When $\alpha = \frac{1}{2}$, the surplus is supermodular, implying

NS-PAM. Analogous calculations yield the same optimal unmatched interval. Figure 1 shows the matching results for $\alpha = 1/2$ (left) and $\alpha = 2$ (right).

To solve for the equilibrium value functions, normalize the outside option of an unmatched column to zero. Under perfect transferability, $v(\delta) = \max_\beta |\beta - \delta|^\alpha - u(\beta)$. Applying Corollary 1 gives $v'(\delta) = \phi_2\big(\mu^{-1}(\delta), \delta\big)$. Together with the boundary condition $v(\delta) = 0$ on $[1/4, 3/4]$ and the equilibrium payoff condition $u(\beta) = |\beta - \mu(\delta)|^\alpha - v\big(\mu(\beta)\big)$, we recover the equilibrium values depicted in Figure 3. Regardless of the matching pattern, the value function is decreasing in type for agents who match upward and increasing in type for agents who match downward. As a result, extreme types have the highest payoffs.

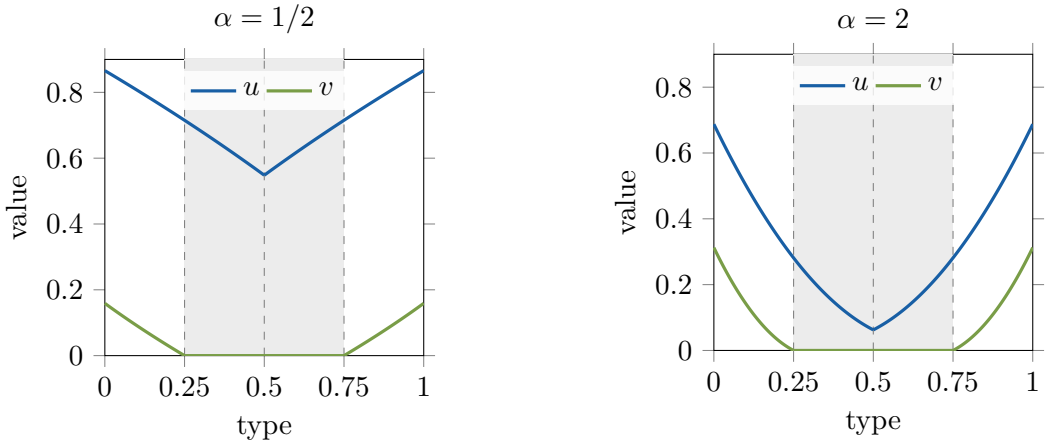


Figure 3: Equilibrium value functions in the transferable-utility uniform example. Blue plots the row value $u(\beta)$ and green plots the column value $v(\delta)$. The gray interval is the unmatched column set $[1/4, 3/4]$, on which $v = 0$; the middle dashed line marks $\beta^* = 1/2$.

D. A Preference for Diversity Implies NAM for NTU Models. In perfectly non-transferable utility (NTU) models, a match between row type β and column type δ yields payoff $U(\beta, \delta)$ to the row and $V(\delta, \beta)$ to the column. Assuming each payoff function weakly exceeds the outside option, a matching is an equilibrium if and only if it is stable, i.e. no unmatched pairs can get together and both become strictly better off. Formally, the degenerate Pareto frontier does not fit our ITU framework. Nonetheless, if we define preferences for diversity in the natural way; namely that U and V obey (5), with the roles of β and δ swapped for V, then we can extend our results to NTU models.[5]

[5]As shown in Legros and Newman (2007a), the behavior of your payoff in your own type does not matter for stable *sorting* patterns in an NTU model.

Proposition 4 *Assume an NTU model with payoffs increasing in diversity. Stable matchings obey NAM, while values $U(\beta, \mu_N(\beta))$ and $V(\delta, \mu_N^{-1}(\delta))$ obey the conclusion of Proposition 3.*

Step 1: The matching must be negative quadrant supported. To extend Proposition 1 to the NTU environment, we prove Step 2 of the proof, namely, there exists a threshold type β^* s.t. $\beta < \beta^*$ matches up and $\beta > \beta^*$ matches down; the remaining steps follow analogously. Toward a contradiction, suppose $\delta \leq \beta < \beta' \leq \delta'$ are matched as (β, δ) and (β', δ'). Preferences for diversity imply $U(\beta, \delta) < U(\beta', \delta)$ and $U(\beta', \delta') < U(\beta, \delta')$. The two inequalities can be connected through stability. Since (β', δ') are matched and $V(\delta', \beta) > V(\delta', \beta')$ by preferences for diversity, stability requires $U(\beta', \delta) \leq U(\beta', \delta')$. Hence, $U(\beta, \delta) < U(\beta, \delta')$, so β and δ' mutually prefer each other to their current partners, contradicting stability.

Step 2: NAM Obtains. By Step 1, any equilibrium matching must be negative quadrant supported. It therefore suffices to show that the matching is NAM within each matched quadrant. Suppose, toward a contradiction, that a stable matching contains positively ordered pairs (β, δ) and (β', δ') with $\beta < \beta'$ and $\delta < \delta'$. In the SE quadrant, both row types match down, thus $\beta' \geq \delta' > \delta$ and $\beta' > \beta \geq \delta$. Since payoffs increase as the types are further apart, $U(\beta', \delta') < U(\beta', \delta)$ and $V(\delta, \beta) < V(\delta, \beta')$. Hence (β', δ) forms a blocking pair, contradicting stability. The argument for the NW quadrant is symmetric.

Step 3: Equilibrium Values. Consider the SE quadrant with $\beta' > \beta > \beta^*$. Since the matching is NAM, we have $\beta > \mu_N(\beta) > \mu_N(\beta')$. Since values are increasing in diversity:

$$U(\beta', \mu_N(\beta')) > U(\beta', \mu_N(\beta)) > U(\beta, \mu_N(\beta))$$

Symmetric logic yields the remaining value orderings in Proposition 3. □

To see how diversity preferences lead to NAM under NTU, note that the row and column types with the largest feasible type gap are mutual ideal partners and therefore must match in equilibrium. For example, consider the case in which the type supports overlap with $\underline{\beta} < \underline{\delta}$ and $\bar{\beta} < \bar{\delta}$. Then $(\underline{\beta}, \bar{\delta})$ is the pair with the largest type gap and must match. By the same reasoning, after removing this pair, the lowest remaining row must match to the highest remaining column, and so on, until the pair with the largest gap switches to the opposite extremes, namely the highest row type and the lowest column type. The same recursive logic applies. Thus, we continue to form NAM pairings until we run out of columns to match.[6]

We turn to sufficient conditions for monotone sorting in ITU models in the next section.

[6]Legros and Newman (2010) prove that a co-ranking condition is necessary and sufficient for NAM in NTU models. We prove in Appendix A.2 that preferences for diversity imply this co-ranking condition.

4 Monotone Sorting

In this section, we discuss and extend existing results on sufficient conditions for assortative matching. Two equilibrium allocations are *payoff equivalent* if their payoff functions differ only on a set of measure zero. Following Legros and Newman (2007a), *the economy exhibits NAM (PAM)* if: (i) there exists an equilibrium in which the matching is NAM (PAM) and (ii) every equilibrium allocation is payoff equivalent to one in which the matching is NAM (PAM). In what follows, we focus on NAM, with analogous arguments applying to PAM.

A. Sufficient Conditions for NAM and NS-PAM. Legros and Newman (2007a) (finite types) and Legros and Newman (2007b) (continuum types) show that *generalized decreasing differences (GDD)* is sufficient for the economy to exhibit NAM:

$$\phi(\beta', \delta', \psi(\delta', \beta, \nu)) \leq \phi(\beta', \delta, \psi(\delta, \beta, \nu)) \quad \forall \beta < \beta',\ \delta < \delta',\ \text{and feasible } \nu$$

The left-hand side is δ''s willingness to pay for β', given that the alternative is a match with β who earns utility ν. The right-hand side is the counterpart expression for δ. Intuitively, this says that a low type δ can weakly outbid a high type δ' in the competition for a higher-type partner.[7] If the inequality is strict, a matching pattern with (β, δ) and (β', δ') cannot be stable: δ will be happy to offer more to β' than the later could get with δ'. Hence NAM obtains as the unique stable matching. When GDD holds weakly, other stable matchings may exist, but Legros and Newman show that they are payoff-equivalent to a NAM equilibrium.

Chade, Eeckhout, and Smith (2017) further show that (strict) GDD is equivalent to *(strictly) single crossing differences (SCD)*; namely: [8]

$$\phi(\beta, \delta', w') \leq \phi(\beta, \delta, w) \quad \Rightarrow \quad \phi(\beta', \delta', w') \leq\ (<)\ \phi(\beta', \delta, w) \quad \forall \beta < \beta' \ \text{ and } \ \delta < \delta' \qquad (11)$$

By Milgrom and Shannon (1994) Theorem 3, if ϕ is twice continuously differentiable and

[7] The same reasoning can be made in terms of β and β' competing for δ'. The GDD condition in this case is $\psi(\delta', \beta', \phi(\beta', \delta, w)) \leq \psi(\delta', \beta, \phi(\beta, \delta, w))$. Using the definition that ψ is the partial inverse of ϕ and $\phi_3 < 0$, one can show that the two conditions are equivalent.

[8] The NAM uniqueness result in Legros and Newman follows from similar logic to that in Step 2 in the proof of Proposition 1. Under strict *single crossing differences*, suppose an equilibrium matches (β, δ) and (β', δ'), with $\beta' > \beta$ and $\delta' > \delta$. Applying strict SCD with $w = v(\delta)$ and $w' = v(\delta')$ yields

$$\phi(\beta, \delta', v(\delta')) \leq \phi(\beta, \delta, v(\delta)) \implies \phi(\beta', \delta', v(\delta')) < \phi(\beta', \delta, v(\delta)).$$

Thus, δ' cannot solve (4) for β', contradicting β' matched to δ'. The same logic shows that weak SCD implies payoff equivalence to NAM. With weak inequalities, both β and β' may be indifferent between δ and δ' at their equilibrium payoffs. Swapping their matches therefore preserves the payoffs of all four agents.

$\phi_3 \neq 0$, single crossing differences is equivalent to the *marginal rate of substitution (MRS)*, $-\phi_2/\phi_3$, being non-increasing in β. In words, a lower type β is willing to pay more for an increment in the partner's type δ. However, the global C^2 assumption on ϕ is not met in our applications, as the Pareto Frontier in our applications is only C^1 when $\beta \neq \delta$. We therefore prove the following lemma in Appendix A.3, which applies to any Pareto frontier, not just those that display a preference for diversity.

Lemma 2 *Assume ϕ is continuous on a compact set $B \subseteq \mathbb{R}^3$ and C^1 on the interior of B with $\phi_3 < 0$. If the MRS $-\phi_2/\phi_3$ is (decreasing) non-increasing in β on the interior of B, then ϕ obeys (strictly) single crossing differences on B.*

If ϕ is twice differentiable, then differentiating $-\phi_2/\phi_3$ shows that (strict) SCD holds when

$$D(\beta, \delta, w) \equiv \phi_2(\beta, \delta, w)\phi_{13}(\beta, \delta, w) - \phi_3(\beta, \delta, w)\phi_{12}(\beta, \delta, w) \leq (<)\ 0 \tag{12}$$

Intuitively, when does inequality (12) hold? We know that $\phi_3 < 0$ and, by (5), the sign of ϕ_2 is equal to the sign of $(\delta - \beta)$. Thus, to sign D term by term, we would need ϕ_{13} to have the same sign as $(\beta - \delta)$ and $\phi_{12} \leq 0$. In words, we need (i) it is less costly for pairs with greater type diversity to transfer utility ($\phi_{13} > (<)\ 0$ when $\beta > (<)\ \delta$), and (ii) types are substitutes in production ($\phi_{12} < 0$). Suppose two row types compete for a lower-type column player. Condition (i) implies that the higher-type row player faces a lower cost of transferring utility to the column player. Condition (ii) implies that the higher-type row player is more productive at the margin in such match. Taken together, these forces ensure that the higher-type row player can outbid the lower-type row player for the low-type column player, as being both more productive and having lower transfer costs.

Lemma 3 *The economy exhibits NAM when* (12) *holds for all $\beta \neq \delta$. If the inequality holds strictly, NAM is the unique equilibrium matching. The reverse inequality provides analogous sufficient conditions for the economy to exhibit NS-PAM (uniquely when strict).*

In the TU case, $\phi(\beta, \delta, w) = g(\beta, \delta) - w$, so $\phi_{13} = 0$. Thus, since $\phi_3 < 0$, condition (12) reduces to $\phi_{12} \leq 0$, which is the familiar submodularity condition from Becker (1973).

Lemma 3 is the Legros and Newman condition restricted to $\beta \neq \delta$ (with Lemma 2 providing some technical assistance). Preferences for diversity play a subtle role in the analysis. First, preferences for diversity are incompatible with the SCD condition for PAM whenever the 45-degree diagonal lies in the interior of the matching support. To see this, consider any

$\beta = \delta < \delta' = \beta'$ such that (β, δ') and (β', δ) are also in the matching support. For $w = w'$, preferences for diversity imply:

$$\phi(\beta, \delta', w') > \phi(\beta, \delta, w) \quad \text{and} \quad \phi(\beta', \delta', w') < \phi(\beta', \delta, w).$$

Nevertheless, PAM within each matched quadrant remains possible, since preferences for diversity imply that the support for equilibrium matchings cannot contain any segment of the 45 degree line (Proposition 1). Hence, the failure of SCD at the diagonal does not preclude positive sorting within each matched quadrant (NS-PAM). In contrast, a matching that is negative quadrant supported and negatively sorted within each matched quadrant is global NAM. Overall, preferences for diversity determine where equilibrium matches can lie, while local SCD conditions determine how types sort within those regions.

Next we explore the equilibrium covariance in values across matched pairs. Let (Q_1, Q_2) be the two quadrants supporting a matching. Then we can decompose the covariance as:

$$\begin{aligned} \sigma_{uv} &\equiv \int_{\underline{\beta}}^{\bar{\beta}} u(\beta) v(\mu(\beta)) dH_R(\beta) - \int_{\underline{\beta}}^{\bar{\beta}} u(\beta) dH_R(\beta) \int_{\underline{\beta}}^{\bar{\beta}} v(\mu(\beta)) dH_R(\beta) \\ &= \mathrm{E}\left[\mathrm{Cov}(u, v)|Q)\right] + \mathrm{Cov}\left(\mathrm{E}[u|Q], \mathrm{E}[v|Q]\right) \end{aligned}$$

where the first term is the within quadrant covariance and the second term is the between quadrant covariance. While the sum of these terms can be positive or negative under both NAM and NS-PAM, the sign of the first term follows from Proposition 3. In particular, given NAM $u(\beta)$ and $v(\mu(\beta))$ are co-monotone in β, while PAM implies that $u(\beta)$ and $v(\mu(\beta)$ must have opposite monotonicity in β. Thus, we have the following corollary.

Corollary 2 *The within quadrant covariance is strictly positive given inequality* (12)*, and strictly negative under the reverse inequality.*

B. Value Uniqueness. The interval of unmatched columns is unique by Proposition 2; and so, the cdf over matched columns types $\hat{H}_C$ is simply the cdf over column types restricted to $[\underline{\delta}, \delta_L^*]$ and $[\delta_H^*, \bar{\delta}]$ scaled by K^{-1}. Together with the row distribution, equilibrium NAM and NS-PAM matching are the unique solution to the market clearing condition (2). NAM and NS-PAM matching are both strictly monotone within each matched quadrant. And for any strictly monotone matching μ_S, the inverse u_S^{-1} is well-defined on each matched-column interval. Since equilibrium value functions are almost everywhere differentiable, the FOC for

the row-side incentive-compatibility (3) holds almost everywhere, i.e.

$$\phi_2(\beta, \mu_S(\beta), v(\mu_S(\beta))) + \phi_3(\beta, \mu_S(\beta), v(\mu_S(\beta)))v'(\mu_S(\beta)) = 0.$$

Changing variables from β to δ gives the differential equation for the value function v

$$v'(\delta) = -\frac{\phi_2(\mu_S^{-1}(\delta), \delta, v(\delta))}{\phi_3(\mu_S^{-1}(\delta), \delta, v(\delta))} \tag{13}$$

Finally, since unmatched columns get outside option v_0, and all equilibria are payoff equivalent to NAM under (12) and to NS-PAM under its inverse, we have the following Corollary.[9]

Corollary 3 *Suppose $K > 1$. If (12) holds, set $\mu_S = \mu_N$; if its reverse holds, set $\mu_S = \mu_{NP}$. In either case, equilibrium value functions are unique. The column value function equals v_0 on $[\delta_L^*, \delta_H^*]$ and is the solution to (13) on $[\underline{\delta}, \delta_L^*]$ with boundary condition $v(\delta_L^*) = v_0$, and on $[\delta_H^*, \bar{\delta}]$ with boundary condition $v(\delta_H^*) = v_0$. The row value is $u(\beta) = \phi\big(\beta, \mu_S(\beta), v(\mu_S(\beta))\big)$.*

We can further complete the characterization of the unmatched interval when NAM is an equilibrium[10]. Proposition 2 established that the interval of unmatched columns is unique and lies at a corner in the two cases: $\phi_H(\underline{\beta}) \leq \phi_L(\underline{\beta})$ and $\phi_H(\bar{\beta}) \geq \phi_L(\bar{\beta})$. Since the payoff gap $\phi_H(\beta) - \phi_L(\beta)$ is strictly downcrossing (Lemma 1), there is only one remaining case: $\phi_H(\underline{\beta}) > \phi_L(\underline{\beta})$ and $\phi_H(\bar{\beta}) < \phi_L(\bar{\beta})$. The three cases are exhaustive and mutually exclusive.

Lemma 4 *Consider an equilibrium with NAM. If $\phi_H(\underline{\beta}) > \phi_L(\underline{\beta})$ and $\phi_H(\bar{\beta}) < \phi_L(\bar{\beta})$, then the unmatched set lies in the interior, and the threshold types $(\beta^*, \delta_L^*, \delta_H^*)$ are the unique solution to the market clearing condition (7) and $\phi_H(\beta^*) = \phi_L(\beta^*)$.*

Proof: We claim that cutoff type β^* must solve $\phi_H(\beta^*) = \phi_L(\beta^*)$, i.e. be indifferent between δ_H^* and δ_L^*. Assume a NAM equilibrium with $\phi_H(\beta^*) > \phi_L(\beta^*)$, then by continuity, all types near β^* would strictly prefer matching to column types near $\delta_H(\beta^*)$, over matching to column types near $\delta_L(\beta^*)$, but NAM requires types $\beta > \beta^*$ near β to match to types near $\delta_L(\beta^*)$. Likewise, if $\phi_H(\beta^*) < \phi_L(\beta^*)$, then we cannot have NAM with types $\beta < \beta^*$ matching to types

[9] In our model, row and column players are drawn from separate populations. In the *common pool model*, types are drawn from a common distribution and the payoff frontier is symmetric. In this alternative model, values are unique given NAM or NS-PAM. To see this, first note that the cutoff type $\beta^* = \delta_L^* = \delta_H^*$ must equal the median type. Under both NAM and NS-PAM, the median type must match with her clone, and each receives the payoff $\hat{v}$ at the intersection of the Pareto frontier and the 45-degree line (i.e. the symmetric Nash-bargaining payoff). The equilibrium value function $u^* = v^*$ is then the unique solution to the differential equation (13) with $v(\hat{\beta}) = \hat{v}$.

[10] A similar characterization holds under NS-PAM, but is less clean.

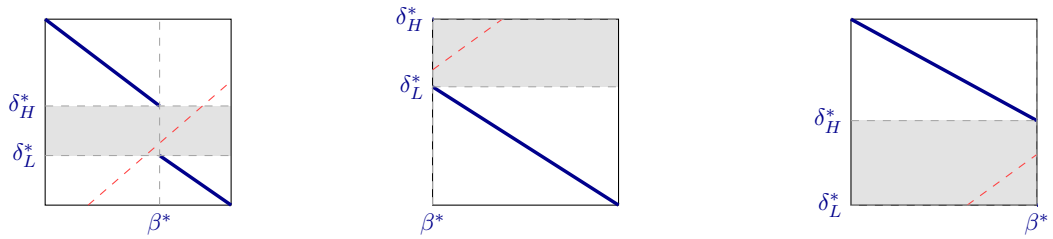


Figure 4: The three unmatched-interval cases and their NAM patterns. The gray band is the unmatched set, and $\delta = \beta$ is the red dashed 45 degree line.

$\delta > \delta_H(\beta^*)$ near $\delta_H(\beta^*)$. Finally, since all equilibria are payoff equivalent to some equilibrium with NAM, type β^* must be indifferent between $\delta_L(\beta^*)$ and $\delta_H(\beta^*)$ in all equilibria.

Now, by Lemma 1 and continuity of ϕ, if $\phi_H(\underline{\beta}) > \phi_L(\underline{\beta})$ and $\phi_H(\bar{\beta}) < \phi_L(\bar{\beta})$, then there exists a unique $\beta^* \in (\underline{\beta}, \bar{\beta})$ such that $\phi_H(\beta^*) = \phi_L(\beta^*)$. Given this unique β^*, the cutoff column types follow from the two market clearing conditions in (7). □

C. The Diversity Index Case. In both of our applications, types interact through an index function, i.e. the Pareto frontier is of the form

$$\phi(\beta, \delta, w) = f(r(\beta, \delta), w), \tag{14}$$

where $r(\beta, \delta)$ is a cardinal measures of type diversity and $f_1(r, w) > 0$. Payoffs increase in diversity when r increases as the two types move farther apart:

$$\beta > \delta \;\Rightarrow\; r_1(\beta, \delta) > 0 > r_2(\beta, \delta) \quad \text{and} \quad \beta < \delta \;\Rightarrow\; r_1(\beta, \delta) < 0 < r_2(\beta, \delta). \tag{15}$$

Hence, $r_1 r_2 < 0$ for all $\beta \neq \delta$. When r and f are twice differentiable, the chain rule yields

$$\phi_1 = f_1 r_1,\;\; \phi_2 = f_1 r_2,\;\; \phi_3 = f_2,\;\; \phi_{13} = f_{12} r_1,\;\; \phi_{23} = f_{12} r_2,\;\; \phi_{12} = f_{11} r_1 r_2 + f_1 r_{12}. \tag{16}$$

Substituting these derivatives 16 into (12) and applying Lemma 3 yields the following sufficient condition for NAM (with the reverse inequality implying NS-PAM):

$$D = r_1 r_2 (f_1 f_{12} - f_2 f_{11}) - f_2 f_1 r_{12} \leq 0 \quad \forall \beta \neq \delta. \tag{17}$$

5 Matching to Play Repeated Games

In this section we assume that payoffs are generated by the subgame perfect equilibria of an infinitely repeated stage game with finite action sets for both players. Assume a public randomization device, so that the set of feasible stage-game payoffs is a convex polygon. The *individually rational (minmax) payoff* π_R (π_C) is the payoff that the row (column) player can secure when the column (row) player tries to minimize the row players payoff, regardless of their own payoff. Let B_R (B_C) be the set of feasible stage-game payoffs in which the row (column) player gets at least π_R (π_C). Thus, $B_R \cap B_C$ is the set of feasible, individually rational, stage-game payoffs. Let $\underline{x}_R$ and $\underline{x}_C$ be the smallest payoff for the row and column (resp.) players on the frontier of $B_R \cap B_C$. We assume that $(u_0, v_0) \in (B_R \cap B_C)^\circ$ (i.e. there exist feasible, individually rational, stage-game payoffs that exceed both players' outside options), and that $(u_0, v_0) > (\underline{x}_R, \underline{x}_C)$ (i.e. that outside options are strictly higher than both the minmax payoff and the smallest payoff on the stage-game Pareto frontier).

The time between periods in the infinitely repeated game is $\Delta > 0$ so that, a row (column) player with discount factors β (δ) discounts next periods payoff by a factor of β^Δ (δ^Δ). Let $V^\Delta(\beta, \delta) \subset \mathbb{R}^2$ be the set of subgame perfect equilibrium (SPE) payoffs for the (row,column) player in this infinitely repeated game. Use $\phi^\Delta(\beta, \delta, w)$ to denote the maximum SPE payoff to a row player of type β when matched to a column player of type δ, assuming that the column player gets SPE payoff w. Naturally, ϕ^Δ is strictly decreasing in w. As a result, it admits a well-defined *partial inverse* ψ^Δ satisfying $\phi^\Delta(\beta, \delta, \psi^\Delta(\delta, \beta, \nu)) \equiv \nu$.

When the two players have identical discount factors, the feasible payoff set of the repeated game coincides with the convex hull of the stage-game payoffs. Moreover, by the Folk Theorem, the SPE payoff set approaches the set of individually rational and feasible stage-game payoffs as players become infinitely patient. With differential discount factors, however, players can achieve payoffs beyond the stage-game payoff frontier.

A. The Continuous-Time Frontier. Lehrer and Pauzner (1999) show that the limit frontier $\phi \equiv \lim_{\Delta \to 0} \phi^\Delta$ for $\beta \neq \delta$ is well defined and provide an explicit formula (their Proposition 2). Toward this characterization, fix any matched pair (β, δ), and let $B(\beta, \delta)$ be the set of feasible stage-game payoffs that weakly exceed the individually rational payoff of the less patient player. Thus, if $\beta > \delta$, $B(\beta, \delta) = B_C$; if $\beta < \delta$, $B(\beta, \delta) = B_R$. The Pareto frontier of $B(\beta, \delta)$ is piecewise-linear with vertices $(x_0, y_0), (x_1, y_1), \ldots, (x_\ell, y_\ell)$, indexed so that $x_0 > \cdots > x_\ell$ (and thus $y_0 < \cdots < y_\ell$). The x-coordinate *always* represents the payoff of the less patient player, while the y-coordinate represents the payoff of the more patient

player. If we want to emphasize the identity of the less patient player, we write (x_i^R, y_i^C) if the row player is weakly less patient and (x_i^C, y_i^R) if the column player is less patient. We say outside options are *high* when $u_0 \geq x_{\ell-1}^R$ and $v_0 \geq x_{\ell-1}^C$.

For each $m = 0, \ldots, \ell - 1$, define the following ratio and sum:

$$r(\beta, \delta) \equiv \frac{\log(\min\{\beta, \delta\})}{\log(\max\{\beta, \delta\})} > 1 \quad \text{and} \quad S_m(r) \equiv \sum_{i=m}^{\ell-1} (x_i - x_{i+1}) \left(\frac{y_{i+1} - y_i}{x_i - x_{i+1}} \right)^{\frac{r}{r-1}} > 0$$

And the sequence of ℓ ordered scalars:

$$\omega_0 \equiv x_0, \qquad \omega_{m+1} \equiv x_m - S_m(r) \left(\frac{y_{m+1} - y_m}{x_m - x_{m+1}} \right)^{\frac{r}{1-r}} \tag{18}$$

As well as the following piecewise function:

$$f(r, w) \equiv y_m + (x_m - w)^{\frac{1}{r}} S_m(r)^{\frac{r-1}{r}} \quad \text{for} \quad \omega_m \geq w \geq \omega_{m+1} \quad \text{and} \quad f(r, w) \geq \pi_p \tag{19}$$

where $\pi_p = \pi_R$ if $\beta > \delta$ and $\pi_p = \pi_C$ if $\beta < \delta$. Note that f is continuous in w by construction. Moreover, since f is strictly decreasing in its second argument, its partial inverse $\hat{f}(r, f(r, w)) = w$ is well defined.

Lemma 5 (Lehrer and Pauzner (1999)) *As $\Delta \to 0$, the Pareto frontier ϕ^Δ of the row player converges uniformly (from below) to the limit Pareto frontier:*

$$\phi(\beta, \delta, w) \equiv \begin{cases} f(r(\beta, \delta), w) & \beta > \delta, \\ \hat{f}(r(\beta, \delta), w) & \beta < \delta. \end{cases}$$

and the set of SPE payoffs $V^\Delta(\beta, \delta) \to V(\beta, \delta) \supseteq B_R \cap B_C$, with strict set inclusion for $\beta \neq \delta$.

In words, f is the limit Pareto frontier when row is more patient than column, while $\hat{f}$ is the Pareto frontier when column is more patient, reflecting the interchange of the two axes.

To fix ideas, we construct the Pareto frontier for the following stage game:

	L	M	R
A	$(1, 6)$	$(0, 0)$	$(0, 0)$
B	$(4, 5)$	$(1, 1)$	$(0, 0)$
C	$(10, 2)$	$(1, 1)$	$(0, 0)$
D	$(11, 1)$	$(1, 1)$	$(0, 0)$

(20)

The minmax payoffs are $\pi_R = 0$ and $\pi_C = 1$. Consider a match in which $\beta > \delta$, so that

the payoffs for the more patient row player are placed on the vertical axis. To construct the Pareto frontier of the SPE payoffs, we first remove all stage game payoffs below the less patient player's IR payoff, $\pi_C = 1$, which removes no payoffs in this example. The remaining vertices of the stage-game frontier (shown in the left panel of Figure 5) are

$$(x_3, y_3) = (1, 11), \qquad (x_2, y_2) = (2, 10), \qquad (x_1, y_1) = (5, 4), \qquad (x_0, y_0) = (6, 1).$$

The slopes of the three segments on the frontier of stage game payoffs are

$$\lambda_2 = \frac{y_3 - y_2}{x_2 - x_3} = 1, \qquad \lambda_1 = \frac{y_2 - y_1}{x_1 - x_2} = 2, \qquad \lambda_0 = \frac{y_1 - y_0}{x_0 - x_1} = 3.$$

Next, define the sums

$$S_2(r) = 1, \qquad S_1(r) = 3 \cdot 2^{\frac{r}{r-1}} + 1, \qquad S_0(r) = 3^{\frac{r}{r-1}} + 3 \cdot 2^{\frac{r}{r-1}} + 1.$$

The corresponding cutoffs separating adjacent frontier segments are

$$\omega_0 = x_0 = 6, \qquad \omega_1 = 6 - \big(3^\rho + 3 \cdot 2^\rho + 1\big)3^{-\rho}, \qquad \omega_2 = 5 - \big(3 \cdot 2^\rho + 1\big)2^{-\rho}, \qquad \omega_3 = 2 - 1 = 1.$$

Since the resulting frontier is everywhere above the row player's individually rational payoff, $\pi_R = 0$, no further truncation is required, and $\phi = f$ is given by (19) with $m = 0, 1, 2$. For example, if $r = \frac{\log \delta}{\log \beta} = 2$, then row's payoff as a function of columns payoff w is

$$\phi(w) = \begin{cases} 1 + \sqrt{22(6 - w)}, & 32/9 \leq w \leq 6, \\ 4 + \sqrt{13(5 - w)}, & 7/4 \leq w \leq 32/9, \\ 10 + \sqrt{2 - w}, & 1 \leq w \leq 7/4. \end{cases}$$

as shown in Figure 5 (right). Notice that the SPE frontier with $r = 2$ (black) dominates the stage-game payoff frontier (blue), which coincides with the feasible payoff frontier of the repeated game when $\beta = \delta$. More generally, prove the following Lemma in Appendix A.4.

Lemma 6 *The limit frontier ϕ obeys all of the assumptions of the general model in Section 2 and payoffs increasing in diversity.*

An immediate Corollary of Lemma 6 is that any equilibrium matching must be negative quadrant supported by Proposition 1 and have quasiconvex values obeying Proposition 3.

Intuitively, in repeated games, larger gaps in discount factors reduce the "congestion" in

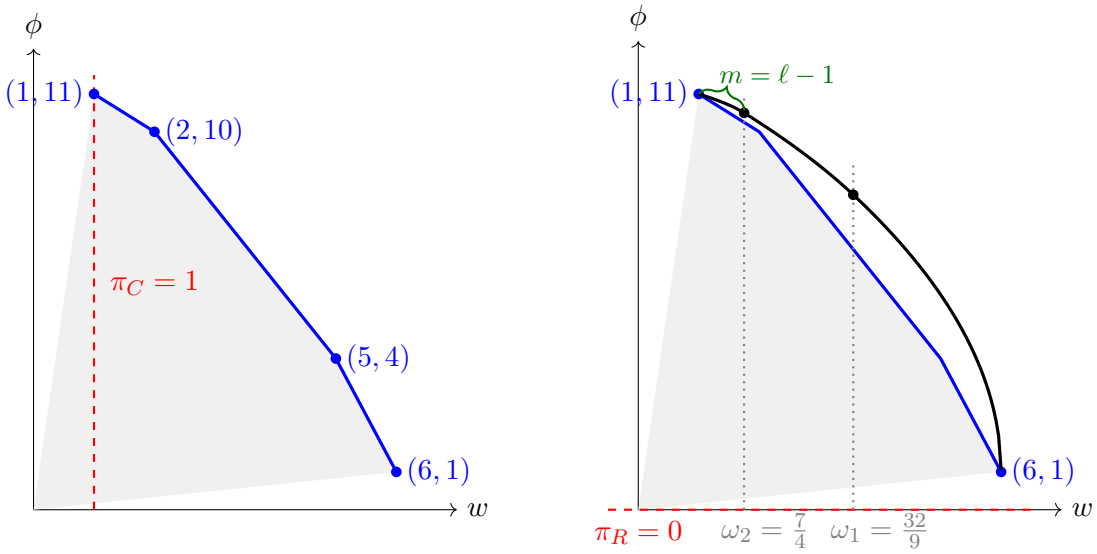


Figure 5: Left: The feasible set and Pareto frontier of stage-game payoffs for game (20). Right: Limit Pareto frontier ϕ (black) of SPE payoffs for $r = 2$.

players' intertemporal interests: the impatient player values earlier rewards relatively more, while the patient player values later rewards relatively more. This creates more room for mutually beneficial intertemporal trade, pushing out the Pareto frontier of feasible repeated-game payoffs and also the Pareto frontier of SPE payoffs. In particular, consider a simple SPE path in which the matched pair receives either (h, l) or (l, h) in each period, with $h > l$. A matched pair has an incentive to front-load the impatient player's reward: play actions yielding (l, h) for (patient, impatient) players up to some period $\bar{t}$, and then switch to actions yielding (h, l) thereafter. Now fix any promised discounted payoff w for the impatient player. Since the patient player prefers an *earlier* switch (smaller $\bar{t}$), she would prefer a lower type.[11]

B. Negative Assortative Matching Given the Limit Frontier. We now explore the sorting patterns at the matching stage given the limiting Pareto frontier. Since ϕ is everywhere continuous and C^1 for all $\beta \neq \delta$ as shown in Lemma 6, Lemma 2 applies. In particular, ϕ obeys SCD if the MRS $-\phi_2/\phi_3$ is non-increasing in β on each segment $w \in (\omega_m, \omega_{m+1})$. That is, ϕ must satisfy inequality (12) on each segment.

We know that $\phi_3 < 0$ and the sign of ϕ_2 is the same as $(\delta - \beta)$ by (5). However, the signs of ϕ_{13} and ϕ_{12} are both ambiguous. The ambiguity in ϕ_{13} can be seen in the right panel of Figure 5. For low values of w, ϕ is flatter than the stage game payoff frontier (so ϕ_3 rises in

[11] The impatient player δ's payoff: $w = (1-\delta)\left(\sum_{k=0}^{\bar{t}-1} \delta^k h + \sum_{k=\bar{t}}^{\infty} \delta^k l\right) = (1-\delta^{\bar{t}})h + \delta^{\bar{t}} l$. Thus the switching period t is pinned down by $\delta^{\bar{t}} = \frac{h-w}{h-l}$. Thus $\bar{t}$ increases in δ.

β), while at high values of w the opposite is true. In this example, ϕ_{12} is also ambiguous: when $r = 2$, ϕ_{12} is strictly positive at $w = 3/2$ and strictly negative at $w = 11/2$. Despite the ambiguity in these terms, we prove in Appendix A.5 that the equivalent condition (17) holds, so that the MRS $-\phi_2/\phi_3$ is non-increasing in β; and so, by Lemma 3, we obtain the following sorting result for *any two-player finite-action stage game*.

Proposition 5 *The economy exhibits NAM given the limit Pareto frontier ϕ. NAM is the unique equilibrium matching for high outside options, $u_0 \geq x^R_{\ell-1}$ and $v_0 \geq x^C_{\ell-1}$.*

Recall that *exhibiting* NAM means that there is an equilibrium allocation with NAM and all equilibria are payoff equivalent to NAM. This does not rule out equilibrium matching patterns that are not NAM for any stage game. The reason for this non-determinancy is that the MRS $-\phi_2/\phi_3$ is non-increasing in β, rather than decreasing. Specifically, for any pair (β, δ) the Pareto frontier is defined on $\ell - 1$ segments, and the MRS is constant on the leftmost of these segments (i.e. the segment with the lowest payoffs for the impatient player) and strictly decreasing on all other segments. For example, in the right graph in Figure 5 the MRS is constant for $w \in [1, 1.75]$ and strictly decreasing for $w \in (1.75, 6]$. Intuitively, NAM will be the unique equilibrium matching as long as equilibrium payoffs are never on this lowest segment of the frontier. This will be true when the impatient player's outside option exceeds the threshold value of w_{l-1} defining this leftmost segment. We show that this threshold value of w_{l-1} never exceeds the payoff for the impatient player on the first kink in the stage game frontier, e.g., 2 in the example in Figure 5. Thus, NAM is the unique equilibrium matching pattern when column outside options are at least $x^C_{\ell-1} = 2$ and row outside options are at least $x^R_{\ell-1} = 4$.

Appendix B further shows that the sorting patterns derived under this limiting Pareto frontier are robust to sufficiently frequent interactions. Specifically, as the interval between interactions $\Delta \to 0$, there exists a convergent sequence of equilibrium allocations, and every such sequence converges to a limit that is payoff-equivalent to a NAM equilibrium. Hence, for sufficiently small Δ, equilibrium payoffs are well approximated by those of a NAM equilibrium. Moreover, when outside options are sufficiently high, the continuous-time frontier ϕ admits NAM as the unique equilibrium matching. In this case, payoff equivalence also pins down the matching, implying that equilibrium matchings converge to NAM as $\Delta \to 0$.

We now illustrate our results with stage game payoffs (20) for $K = 6/5$, β uniform on $[2/3, 11/12]$, δ uniform on $[1/2, 4/5]$, $u_0 = 2$, and $v_0 = 3/2$. Using (8) we have endpoints:

$$\delta_H(\underline{\beta}) = \frac{4}{5}, \quad \delta_L(\underline{\beta}) = \frac{3}{4}, \quad \delta_H(\bar{\beta}) = \frac{11}{20}, \quad \delta_L(\bar{\beta}) = \frac{1}{2}.$$

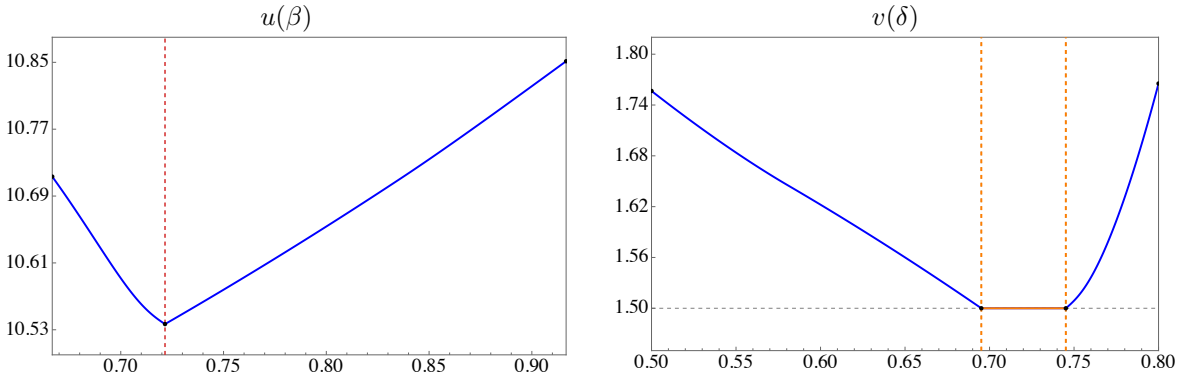


Figure 6: Equilibrium value functions for the numeric example with $K = 6/5$; dashed lines mark the cutoff type β^* and the unmatched column interval (δ_L^*, δ_H^*).

Since $\delta_H(\underline{\beta}) > \delta_L(\underline{\beta}) > \underline{\beta}$, payoffs increasing in diversity implies that $\phi_H(\underline{\beta}) > \phi_L(\underline{\beta})$, and similarly, since $\delta_L(\bar{\beta}) < \delta_H(\bar{\beta}) < \bar{\beta}$, we have $\phi_H(\bar{\beta}) < \phi_L(\bar{\beta})$. Therefore, middle types are unmatched by Lemma 4 with cutoff type β^* solving: $\phi_H(\beta^*) = \phi_L(\beta^*) \Rightarrow \beta^* \approx 0.722$. By market clearing (7), the unmatched column interval is $(\delta_L^*, \delta_H^*) \approx (0.695, 0.745)$, and the equilibrium NAM matching function is

$$\mu_N(\beta) = \begin{cases} \frac{22}{15} - \beta, & \beta \in \left[\frac{2}{3}, \beta^*\right], \\ \frac{17}{12} - \beta, & \beta \in \left[\beta^*, \frac{11}{12}\right]. \end{cases}$$

We can then apply Corollary 3 with this matching function and the unmatched column interval to solve for the unique equilibrium value functions, which are illustrated in Figure 6.

6 Matching Diverse Information Sources

This application is adapted from Chade and Eeckhout (2018), who study the optimal allocation of experts to teams. In their model, each team member observes a separate signal about an unknown state. An agent's type determines the precision of their signal, while the correlation across signals is exogenously fixed and independent of the matching. Signals arrive after matches are formed, at which point signals are shared and the team chooses a common action. This common action determines the match payoff, which is perfectly divisible between agents. Match payoffs fall in the quadratic distance between the chosen action and the state. They derive sufficient conditions for negative sorting by precision.

Their model does not fit our general framework, since match payoffs are globally increas-

ing in signal precision (vertically differentiated types). Instead, we consider a complimentary information aggregation model with horizontally differentiated types. Specifically, all experts have a common signal precision, while the correlation between their signals depends on the proximity of their information sources (types). We then introduce heterogeneous preferences over the optimal team action. This conflict of interest within each team produces an imperfectly transferable utility matching model.

A. Transferable Utility with Horizontal Information Diversity. As in Chade and Eeckhout (2018), there is an unknown state of nature $s \in \mathbb{R}$ with a Gaussian prior $\tilde{s} \sim \mathcal{N}(\mu, \tau^{-1})$, where $\tau > 0$ denotes the prior precision. Once matched, each agent i observes a signal $\tilde{\xi}_i = s + \tilde{\varepsilon}_i$ with $\tilde{\varepsilon}_i \sim \mathcal{N}(0, \lambda^{-1})$ with signal precision $\lambda > 0$.

We assume that an agent's type is the location of their information source, with signal correlation determined by the distance between sources. Specifically, when types β and δ match, the correlation between their signals is given by $\text{corr}(\tilde{\varepsilon}_1, \tilde{\varepsilon}_2) = \rho(|\beta - \delta|)$, where the *correlation profile* $\rho : [0,1] \to [0,1)$ is C^2 and satisfies $\rho'(|\beta - \delta|) < 0$ for all $\beta \neq \delta$. Thus, more distant sources generate more independent, and hence less redundant, signals.

After observing both signals, the matched pair forms the Bayesian posterior over state s and jointly chooses an action $a \in \mathbb{R}$ to maximize $2\left(\pi - \mathbb{E}\left[(a-s)^2\right]\right)$, for constant $\pi > 0$. As shown in Chade and Eeckhout (2018), the pair's maximized expected payoff, denoted by $2V$, has a closed form solution where V is given by:

$$V(|\beta - \delta|) = \pi - \frac{1 + \rho(|\beta - \delta|)}{\tau(1 + \rho(|\beta - \delta|)) + 2\lambda}, \tag{21}$$

where $V'(|\beta - \delta|) > 0$ for $\beta \neq \delta$. Utility is perfectly transferable within the pair, i.e., $\phi(\beta, \delta, w) = 2V(|\beta - \delta|) - w$. Thus, unlike in the original model, payoffs increase in diversity. Intuitively, since signal correlation decreases with the distance between information sources, more diverse pairs generate more informative signals, leading to a higher posterior precision and a larger match value.

By Proposition 1, any equilibrium matching must be negatively quadrant supported. The remaining question is how agents sort *within* each matching quadrant. For $\beta \neq \delta$, the cross-partial of $V(|\beta - \delta|)$ has the opposite sign of the curvature of V:

$$\frac{\partial^2 V(|\beta - \delta|)}{\partial \beta \partial \delta} = -V''(|\beta - \delta|) = \frac{2\lambda}{A(d)^3}\left[A(d)\rho''(d) - 2\tau\rho'(d)^2\right], \tag{22}$$

where $d \equiv |\beta - \delta|$ and $A(d) \equiv \tau\big(1 + \rho(d)\big) + 2\lambda > 0$. Notice that $V'' \leq 0$ cannot hold

globally. Specifically, it fails when zero type difference $d = 0$ is in the interior of the support. If $V'(0) \neq 0$, the cross-partial is not defined at the diagonal; if $V'(0) = 0$ while $V'(|\beta - \delta|) > 0$ for all $\beta \neq \delta$, then V' must be increasing somewhere, implying $V'' > 0$ somewhere. Hence global PAM cannot arise. Nevertheless, $V'' \leq 0$ may hold within a matched quadrant, since the quadrant does not cross the diagonal and therefore excludes zero type differences from its interior. Thus, away from the diagonal, we have:

$$\rho'' < (\rho')^2 \left[\frac{2\tau}{A}\right] \quad \Rightarrow \quad \text{NAM} \qquad \text{and} \qquad \rho'' > (\rho')^2 \left[\frac{2\tau}{A}\right] \quad \Rightarrow \quad \text{NS-PAM} \tag{23}$$

Although the total payoff always increases with diversity, the sorting pattern depends on how the marginal value of diversity changes with distance. If V is concave, the marginal value of additional diversity is diminishing, generating local supermodularity and thus the standard force toward positive assortative matching. Since $\rho'(|\beta - \delta|) < 0$ for all $\beta \neq \delta$, the right-hand side of each inequality in (23) is *strictly* positive, NAM obtains whenever the correlation function ρ is weakly concave in information source diversity, while NS-PAM requires that the correlation function must be sufficiently convex (log-convexity is sufficient).

B. An ITU Extension with A Preference Bias. We now retain the information structure above, and introduce disagreement over the optimal action. Specifically, given state s and action a, the payoffs to the row and column player are

$$u_R(a, s) = \pi - (a - s)^2 \quad \text{and} \quad v_C(a, s) = \pi - (a - s - b)^2, \tag{24}$$

where $b > 0$ is the preference bias common to all column agents. The economic motivation for the bias term is that agents may have different objectives even when they observe the same information. For example, a risk manager and a product manager may evaluate the same information differently: the former may prefer a more conservative action, while the latter may favor a more aggressive one.

Lemma 7 *The Pareto frontier of payoffs from matching information sources is given by*

$$\phi(\beta, \delta, w) = f(d, w) \equiv V(d) - \left[b - \sqrt{V(d) - w}\right]^2, \quad w \in [V(d) - b^2, V(d)], \tag{25}$$

with $d = |\beta - \delta|$ and $V(d)$ defined as in (21)*. Thus, payoffs are increasing in diversity.*

PROOF: We trace out the Pareto frontier by varying $\theta \in [0,1]$ in the following maximization:

$$\max_a \{\theta \mathbb{E}[u(a,s) \mid \xi_1, \xi_2] + (1-\theta)\mathbb{E}[v(a,s) \mid \xi_1, \xi_2]\},$$

The optimal joint action is thus the weighted average: $a_\theta(\xi_1, \xi_2) = \mathbb{E}[s \mid \xi_1, \xi_2] + (1-\theta)b$. The expected losses are equal to

$$\mathbb{E}[(a_\theta - s)^2] = \mathrm{Var}(s \mid \xi_1, \xi_2) + (1-\theta)^2 b^2, \quad \mathbb{E}[(a_\theta - s - b)^2] = \mathrm{Var}(s \mid \xi_1, \xi_2) + \theta^2 b^2.$$

Substituting into (24) and using the definition of $V(d) = \pi - \mathrm{Var}(s \mid \xi_1, \xi_2)$ gives

$$u_R = V(d) - b^2(1-\theta)^2, \qquad v_C = V(d) - b^2\theta^2. \tag{26}$$

Eliminating θ yields $\sqrt{V(d) - u_R} + \sqrt{V(d) - v_C} = b$, which gives the stated frontier (25). As θ ranges over $[0,1]$, v_C ranges from $V(d) - b^2$ to $V(d)$, giving the stated domain. By inspection ϕ is strictly increasing in d. □

We now applying the MRS condition (17) to the Pareto frontier 25 to characterize sorting pattern. Since match diversity is $d(\beta, \delta) = |\beta - \delta|$, we substitute $d_1 d_2 = -1$ and $d_{12} = 0$ into (17) to get $D = f_2 f_{11} - f_1 f_{12}$, which given derivatives (suppressing arguments)

$$f_1 = \frac{bV'}{\sqrt{V-w}}, \quad f_2 = 1 - \frac{f_1}{V'}, \quad f_{12} = \frac{f_1}{2(V-w)}, \quad f_{11} = \frac{bV''}{\sqrt{V-w}} - f_{12}V'.$$

yields

$$D = -\frac{b}{2\left(V(d) - w\right)^{\frac{3}{2}}}\left(V'(d)^2 - 2(V(d) - w - b(V(d)-w)^{\frac{1}{2}})V''(d)\right).$$

Thus, since $b > 0$,

$$D \lesseqgtr 0 \quad \Leftrightarrow \quad V'(d)^2 \gtreqless \left(V(d) - w - b(V(d)-w)^{\frac{1}{2}}\right)V''(d) \equiv g(d,w)V''(d) \tag{27}$$

We show below that $g(d,w) \leq 0$, thus global PAM cannot arise as in the TU case. Altogether, NAM obtains when $(V')^2 \geq gV''$ and NS-PAM obtains for the opposite inequality.

Proposition 6 *Given the matching diverse information sources Pareto frontier* (25)*:*
(*a*) *NAM obtains given the left hand inequality in* (23)*;*
(*b*) *There exists* $\underline{b}$ *such that NAM obtains for all* $b < \underline{b}$*;*
(*c*) *If the right hand inequality in* (23) *holds, then there exists finite* $\bar{b}$ *such that NS-PAM obtains for all* $b > \bar{b}$*.*

PROOF OF (a): Since g is strictly convex in w for all d, with endpoints $g(d, V(d) - b^2) = g(d, V(d)) = 0$ and initial slope $g_w(d, V(d) - b^2) = -1/2$, it is strictly negative on the interior of its domain $w \in [V(d) - b^2, V(d)]$. Consequently, convex V is sufficient for NAM as in the TU sub-section above model, which is equivalent to the left hand inequality in (23).

PROOF OF (b): The function V is independent of b with a strictly positive first derivative. Moreover, The domain for w shrinks to V as $b \to 0$; and thus $g \to 0$ as $b \to 0$. Altogether, $(V')^2$ is strictly positive, while $gV'' \to 0$ as $b \to 0$, and so $D < 0$ for sufficiently small b.

PROOF OF (c): As $b \to \infty$, the domain for payoffs expands to the real line below $V(d)$, and thus outside options must bind for both columns and rows, ensuring w bounded and bounded away from $V(d)$. That is, w lies between the column's outside option and the corresponding column payoff at the row's outside option. Moreover, g converges to $-\infty$ as $b \to \infty$ on any bounded interval away from $V(d)$. Consequently, if V is strictly concave, it follows that gV'' converges to ∞ and thus $(V')^2 < gV''$ on such bounded intervals. As shown in the TU sub-section above, strict concavity of V is equivalent to the right hand inequality in (23). □

Notice that the ITU model subsumes the NTU and TU models as the limiting cases of $b \to 0$ and $b \to \infty$, respectively. More generally, NAM is easier to secure in the ITU model than the TU model. Indeed, the TU NAM condition is sufficient for NAM in the ITU model, while the TU NS-PAM condition is only sufficient in the $b \to \infty$ limit. The intuition is that imperfect transferability introduces an additional force toward NAM. Under TU, a higher-diversity match generates more surplus, which can be redistributed costlessly between partners, so sorting is determined solely by the curvature of the match value function $V(d)$. Under ITU, by contrast, redistributing surplus requires changing the compromise over the joint action. Since $f_{12} > 0$, the marginal cost of transferring utility is lower in matches with greater type diversity. Consequently, a row agent in a more diverse match can compensate a potential partner more cheaply than a row agent in a less diverse match, making it more competitive in attracting that partner. This additional advantage strengthens the force toward NAM.

7 Conclusion

A Omitted Proofs

A.1 Proof for the Uniqueness of the Unmatched Interval

This appendix shows that when columns outnumber rows ($K > 1$), every equilibrium has the same threshold types $(\beta^*, \delta_L^*, \delta_H^*)$, and thus the same set of unmatched columns. We extend the proof for Property 1 in Demange and Gale (1985) to continuum types and bounded payoffs. The result holds for any ITU frontier with payoffs increasing in diversity (5).

We use three facts established in the text. First, the payoffs of every matched pair lie on the Pareto frontier in any equilibrium. Second, by Proposition 1, the unmatched column types form an interval with thresholds $\delta_L^* \leq \delta_H^*$ and H_C-mass $(K-1)/K$. The unmatched distribution is the normalized restriction of H_C to this interval. Thus, for any $J \subseteq [\delta_L^*, \delta_H^*]$,

$$S_C(J) = \frac{K}{K-1} H_C(J);$$

and the cdf over matched columns $\hat{H}_C$ obeys (1). Third, by Proposition 3, equilibrium value functions are strictly monotone on each side of the threshold types.

Step 1 *Let (S_C, M, u, v) and (S'_C, M', u', v') be equilibria. Define*

$$R \equiv \{\beta : u(\beta) > u'(\beta)\},\ R' \equiv \{\beta : u(\beta) < u'(\beta)\},\ R^0 \equiv \{\beta : u(\beta) = u'(\beta)\},$$

and C, C', C^0 analogously for columns. R is matched with C' and R' with C in both equilibria.

Fix $(\beta, \delta) \in \text{supp}(M)$ with $\beta \in R$, so that $u(\beta) = \phi(\beta, \delta, v(\delta))$ by (3). Since $u(\beta) > u'(\beta) \geq u_0 \geq \underline{u}(\beta, \delta)$, and $\phi_3 < 0$, we have $v(\delta) < \bar{v}(\beta, \delta)$. Toward a contradiction, assume $\delta \notin C'$, i.e. $v(\delta) \geq v'(\delta)$. Thus $\bar{v}(\beta, \delta) > v'(\delta) \geq v_0 \geq \underline{v}(\beta, \delta)$, and since $\phi_3 < 0$:

$$\phi(\beta, \delta, v'(\delta)) \geq \phi(\beta, \delta, v(\delta)) = u(\beta) > u'(\beta).$$

But stability of M' requires $u'(\beta) \geq \phi(\beta, \delta, v'(\delta))$, yielding a contradiction. Hence $\delta \in C'$, so M matches R and C'. Symmetrically, choosing $\delta \in C'$ shows that M' also matches R and C'. Analogous arguments for R' and C completes the proof.

Step 2 $S_c(C') = S'_c(C) = 0$.

For any $\delta \in C'$, $v'(\delta) > v(\delta) \geq v_0$, so δ is matched under M' with a row partner in R. Hence

$$K H_C(C') \leq H_R(R). \tag{28}$$

Symmetrically, any $\beta \in R$ is matched under M with a column partner in C'. The available mass of matched columns in C' is $KH_C(C') - (K-1)S_C(C')$, so

$$H_R(R) \leq KH_C(C') - (K-1)S_C(C'). \tag{29}$$

Combining inequalities (28) and (28) yields $S_C(C') = 0$. Similarly, $S'_C(C) = 0$.

Step 3 *Uniqueness of* $(\beta^*, \delta_L^*, \delta_H^*)$ *Across all Equilibria.*

Let (S_C, M, u, v) and (S'_C, M', u', v') be equilibria with unmatched column intervals (δ_L, δ_H) and (δ'_L, δ'_H), respectively. Toward a contradiction, suppose the two intervals differ. Since both unmatched intervals have the same H_C-mass and H_C has full support thus strictly increasing on $[\underline{\delta}, \bar{\delta}]$, we have $\delta_L \neq \delta'_L$. WLOG suppose $\delta_L < \delta'_L$ and $\delta_H < \delta'_H$.

Consider the non-empty open interval $J \equiv (\delta_L, \min\{\delta'_L, \delta_H\})$. For every $\delta \in J$:

- $\delta \in [\delta_L, \delta_H] = \text{supp}(S_C)$, and so $v(\delta) = v_0$;
- $\delta < \delta'_L$, and so Proposition 3(ii) yields $v'(\delta) > v'(\delta'_L) \geq v_0$.

Thus $J \subseteq C'$. Since J lies inside the unmatched interval of the first equilibrium and has positive H_C-mass,

$$S_C(C') \geq S_C(J) = \frac{K}{K-1} H_C(J) > 0,$$

contradicting the mass balance above. Therefore $(\delta_L, \delta_H) = (\delta'_L, \delta'_H)$, and the unmatched distributions coincide. The market clearing condition (7) uniquely determines β^*. □

A.2 Preferences for Diversity Imply Reverse Co-Ranking

Legros and Newman (2010) show a *co-ranking* condition is a necessary and sufficient condition for PAM under NTU; namely that for any two men and two women, either the top man and woman prefer each other, or the bottom man and woman do. Reversing the ordering on one side gives the analogous co-ranking condition for NAM: for every $\beta < \beta'$ and $\delta < \delta'$, either (β, δ') or (β', δ) mutually strictly prefer each other.

Preferences for diversity imply this co-ranking condition. Assume $U(\beta, \delta)$ and $V(\delta, \beta)$ obey (5). Figure 7 displays the six possible orderings given $\beta < \beta'$ and $\delta < \delta'$. In cases (1)–(2), β and δ' mutually prefer each other; in cases (3)–(4), β' and δ do.

In case (5), each column prefers the opposite row: δ prefers β' and δ' prefers β. If β' also prefers δ, co-ranking follows. Otherwise,

$$U(\beta, \delta) < U(\beta', \delta) \leq U(\beta', \delta') < U(\beta, \delta'),$$

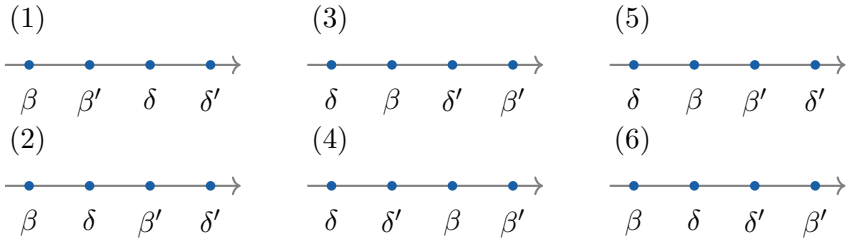


Figure 7: The six orderings of $\beta < \beta'$ and $\delta < \delta'$.

where the first and last inequalities are implied by preferences for diversity. Hence, β prefers δ' so co-ranking also follows. Case (6) is symmetric.

A.3 Proof of Lemma 2

PROOF: Define $h(\beta) \equiv \phi(\beta, \delta', w') - \phi(\beta, \delta, w)$. We show that $h(\beta) = 0 \Rightarrow h(\beta') \leq 0$ (< 0 with a decreasing MRS) for all $\beta' > \beta$ and $\delta' > \delta$. The case $h(\beta) < 0$ follows from continuity.

Step 1 *Constructing an indifference path.*

Suppose $\phi(\beta, \delta', w') = \phi(\beta, \delta, w)$. Since ϕ is C^1 on the interior of B with $\phi_3 < 0$, the Implicit Function Theorem uniquely defines a C^1 function $\gamma : (\delta, \delta') \to \mathbb{R}$ satisfying

$$\phi(\beta, t, \gamma(t)) = \phi(\beta, \delta, w) \quad \forall\, t \in [\delta, \delta'],$$

with $\gamma(\delta) = w$, $\gamma(\delta') = w'$, and

$$\gamma'(t) = -\frac{\phi_2(\beta, t, \gamma(t))}{\phi_3(\beta, t, \gamma(t))}. \tag{30}$$

Step 2 *Evaluating $h(\beta')$ along this path for any $\beta' > \beta$.*

The function γ inherits continuity on B from ϕ; and so integrate along γ for any $\beta' > \beta$:[12]

$$h(\beta') = \phi(\beta', \delta', \gamma(\delta')) - \phi(\beta', \delta, \gamma(\delta)) = \int_{\delta}^{\delta'} \Big[\phi_2(\beta', t, \gamma(t)) + \phi_3(\beta', t, \gamma(t))\, \gamma'(t)\Big]\, dt.$$

Substitute for γ' using (30) and rearrange to get:

$$h(\beta') = \int_{\delta}^{\delta'} \phi_3(\beta', t, \gamma(t)) \left[\frac{\phi_2(\beta', t, \gamma(t))}{\phi_3(\beta', t, \gamma(t))} - \frac{\phi_2(\beta, t, \gamma(t))}{\phi_3(\beta, t, \gamma(t))}\right] dt.$$

[12]Here we use the improper integral when the limits of integration are on the boundary of B.

Since the MRS is non-increasing in β, the term in square brackets is non-negative, while $\phi_3 < 0$. Altogether, the integrand is weakly negative when the MRS is non-increasing. Consequently, $h(\beta') \leq 0$ for all $\beta' > \beta$ and $\delta' > \delta$. By similar logic, the integrand must be strictly negative when the MRS is decreasing; so that, $h(\beta') < 0$ for all $\beta' > \beta$ and $\delta' > \delta$. $\square$

A.4 Proof of Lemma 6

We fix $\beta \geq \delta$, and so $r(\beta, \delta) = \frac{\log(\delta)}{\log(\beta)}$. Analogous steps prove the result for $\beta < \delta$.

Step 1 *Two Useful Identities.*

Define:

$$q_m \equiv \left(\frac{y_m - y_{m-1}}{x_{m-1} - x_m} \right)^{\frac{r}{r-1}}$$

Then by definition of the breakpoint ω_m in (18)

$$x_{m-1} - \omega_m = S_{m-1}(r) q_m(r) \tag{31}$$

By construction, S_m obeys the difference equation:

$$S_{m-1} = (x_{m-1} - x_m) q_m^{-1} + S_m$$

Substituting this into (31) and manipulating we discover:

$$\begin{aligned} x_{m-1} - \omega_m &= \left((x_{m-1} - x_m) q_m^{-1} + S_m \right) q_m = x_{m-1} - x_m + S_m q_m \\ &\Rightarrow \quad x_m - \omega_m = S_m q_m \end{aligned} \tag{32}$$

Step 2 *The limit Pareto frontier is continuously differentiable in β.*

Since $r(\cdot)$ is C^2, ϕ will be continuously differentiable if f_1 and f_2 are continuous at kinks u_m. First consider f_2. For $w \in (\omega_{m+1}, \omega_m)$ we compute:

$$f_2(r, w) = -\frac{1}{r} S_m^{(r-1)/r} (x_m - w)^{1/r - 1}.$$

Taking the limit using (32) gives

$$\lim_{w \uparrow u_m} f_2(r, w) = -\frac{1}{r} q_m^{1/r - 1}.$$

For $w \in (\omega_m, \omega_{m-1})$ we have:

$$f_2(r,w) = -\frac{1}{r} S_{m-1}^{(r-1)/r} (x_{m-1} - w)^{1/r-1}.$$

Taking the limit using (32) yields the same limit

$$\lim_{w \downarrow \omega_m} f_2(r,w) = -\frac{1}{r} q_m^{1/r-1}.$$

Now consider f_1. Differentiating equation (19) yields

$$f_1(r,w) = (x_m - w)^{\frac{1}{r}} S_m^{\frac{r-1}{r}} \left[\frac{\log(S_m) - \log(x_m - w)}{r^2} + \frac{r-1}{r} \frac{\partial S_m / \partial r}{S_m} \right] \tag{33}$$

Evaluating (33) at the boundary $w = \omega_m$ using (32) gives

$$\lim_{w \uparrow \omega_m} f_1(r,w) = q_m^{\frac{1}{r}} S_m \left[-\frac{\log(q_m)}{r^2} + \frac{r-1}{r} \frac{\partial S_m / \partial r}{S_m} \right].$$

A similar computation for $w \in (\omega_m, \omega_{m-1})$ using (31) yields:

$$\lim_{w \downarrow \omega_m} f_1(r,w) = q_m^{\frac{1}{r}} S_{m-1} \left[-\frac{\log(q_m)}{r^2} + \frac{r-1}{r} \frac{\partial S_{m-1} / \partial r}{S_{m-1}} \right].$$

Thus, the difference in the directional limits is:

$$\lim_{w \downarrow \omega_m} f_1(r,w) - \lim_{w \uparrow \omega_m} f_1(r,w) = q_m^{\frac{1}{r}} \left(\frac{r-1}{r} \left(\frac{\partial S_{m-1}}{\partial r} - \frac{\partial S_m}{\partial r} \right) + \frac{\log(q_m)}{r^2} (S_m - S_{m-1}) \right) = 0$$

where the final inequality follows from

$$S_{m-1} - S_m = (x_{m-1} - x_m) \left(\frac{y_m - y_{m-1}}{x_{m-1} - x_m} \right)^{\frac{r}{r-1}}$$

and thus,

$$\frac{\partial S_{m-1}}{\partial r} - \frac{\partial S_m}{\partial r} = (S_{m-1} - S_m) \frac{\log(q_m)}{r(r-1)}.$$

Step 3 *Outside Options.*

Fix any matched pair (β, δ). We have assumed that $(u_0, v_0) \in (B_R \cap B_C)^\circ \subset B_R \cap B_C \subseteq V(\beta, \delta)$; and thus, $u_0 < \phi(\beta, \delta, v_0)$. Furthermore, we have assumed $v_0 > \underline{x}_C$; and thus by construction, $v_0 > \underline{v}(\beta, \delta)$ (the smallest column payoff on ϕ).

Step 4 *Payoffs are increasing in diversity.*

We prove the result for $\beta > \delta$. The argument for $\beta < \delta$ is analogous. When $\beta > \delta$, $\phi_1 > 0$ is equivalent to $f_1(r, w) > 0$ by (15) and (16). And since ϕ is C^1 by Step 2, we need only sign $f_1(r, w) > 0$ on segments $w \in (\omega_{m+1}, \omega_m)$. By (33), the derivative $f_1(r, w)$ has the same sign as

$$\log\left(\frac{S_m}{x_m - w}\right) + r(r-1)\frac{\partial S_m/\partial r}{S_m}.$$

Define $\rho \equiv r/(r-1)$ and $\lambda_m \equiv \frac{y_{m+1}-y_m}{x_m - x_{m+1}} > 0$, so that $S_m = \sum_{i=m}^{l-1}(x_i - x_{i+1})\lambda_i^\rho$, and

$$\frac{\partial S_m}{\partial r} = \frac{dS_m}{d\rho}\frac{d\rho}{dr} = \left(\sum_{i=m}^{l-1}(x_i - x_{i+1})\lambda_i^\rho \log(\lambda_i)\right)\left(-\frac{1}{(r-1)^2}\right).$$

Therefore,

$$r(r-1)\frac{\partial S_m/\partial r}{S_m} = -\rho\frac{\sum_{i=m}^{l-1}(x_i - x_{i+1})\lambda_i^\rho \ln \lambda_i}{\sum_{k=m}^{l-1}(x_k - x_{k+1})\lambda_k^\rho} \equiv -\rho\sum_{i=m}^{l-1}\chi_i \ln \lambda_i.$$

where $\chi_i \equiv \frac{(x_i - x_{i+1})\lambda_i^\rho}{\sum_{k=m}^{l-1}(x_k - x_{k+1})\lambda_k^\rho}$, and thus, $\sum \chi_i = 1$. Then $f_1(r, w) > 0$ is equivalent to

$$\ln\left(\frac{S_m}{x_m - w}\right) > \rho\sum_{i=m}^{l-1}\chi_i \ln \lambda_i \quad \Leftrightarrow \quad x_m - w < S_m(\Pi_{i=m}^{l-1}\lambda_i^{\chi_i})^{-\rho} \tag{34}$$

In the right inequality, the RHS is independent of the promised payoff w. Thus, we need only verify that the inequality holds at the lowest value of $w = \omega_{m+1} \equiv x_m - S_m\lambda_m^{-\rho}$. Evaluating inequality (34) at this value of w and rearranging using $S_m > 0$ and $\rho > 1$ we get:

$$\lambda_m \geq \Pi_{i=m}^{l-1}\lambda_i^{\chi_i}.$$

which holds by the power mean inequality since: (i) the LHS is the maximum λ_i for $i = m, m+1, \ldots, l-1$, and (ii) the RHS is the geometric mean of the λ_i's (since $\sum_{i=m}^{l-1}\chi_i = 1$ by construction). The inequality is strict, except for $m = l-1$, in which case inequality (34) is strict except for one point $w = \omega_{m+1} = \omega_\ell = \underline{v}(\beta, \delta)$. Altogether, $f_1(r, w) > 0$ for all $w \neq \omega_l$. Since $v_0 > \underline{v}(\beta, \delta)$, we conclude $\phi_1(\beta, \delta, w) > 0$ for all $\beta > \delta$. Finally, ϕ_2 has the opposite sign by (15) and (16). □

A.5 Proof of Proposition 5

Step 1 *A sufficient condition for a Decreasing MRS* $-\phi_2/\phi_3$.

By Lemma 6 the ratio $-\phi_2/\phi_3$ is continuous at the kinks ω_m and at $\beta = \delta$. Thus, we only need to show that $-\phi_2/\phi_3$ is non-increasing in β on each segment m for $\beta > \delta$. And since ϕ is C^2 on each segment, this is implied by inequality (17). Substituting the following derivatives

$$r_1 = -\frac{\log(\delta)}{\beta \log(\beta)^2} > 0 > \frac{1}{\delta \log(\beta)} = r_2 \quad \Rightarrow \quad r_{12} = -\frac{1}{\beta\delta \log(\beta)^2} < 0 \quad \text{and} \quad \frac{r_1 r_2}{r_{12}} = r \tag{35}$$

into (17) and rearranging yields:

$$D = r_1 r_2 \left(f_1 f_{12} - f_2 f_{11} - \frac{f_2 f_1}{r} \right).$$

Since, $r_1 r_2 < 0$, the sign of D is equal to the sign of

$$g(r, w) \equiv f_2(r, w) f_{11}(r, w) + \frac{f_2(r, w) f_1(r, w)}{r} - f_1(r, w) f_{12}(r, w)$$

Altogether, $-\phi_2/\phi_3$ will be non-increasing in β if $g \leq 0$ on each segment m, and strictly decreasing in β when $g < 0$.

Step 2 *The MRS $-\phi_2/\phi_3$ is non-increasing in β, and strictly decreasing for $w > \omega_{\ell-1}$.*

Define the slope of the Pareto frontier on each segment:

$$\lambda_i \equiv \frac{y_{i+1} - y_i}{x_i - x_{i+1}} > 0, \qquad i = 0, \ldots, \ell - 1 \quad \Rightarrow \quad \lambda_i > \lambda_{i+1}$$

since the Pareto frontier is concave.

Let $\rho \equiv \frac{r}{r-1} > 1$, $a_i \equiv (x_i - x_{i+1}) > 0$, and $b_i \equiv \log \lambda_i$, so that $S_m(r) = \sum_{i=m}^{\ell-1} a_i e^{\rho b_i}$ and

$$H_m(\rho) \equiv \log S_m\big(r(\rho)\big) = \log \left(\sum_{i=m}^{\ell-1} a_i e^{\rho b_i} \right)$$

Since $a_i e^{\rho b_i}$ is log-linear in ρ and log-convexity is preserved by summation, we have:

$$H_m''(\rho) = 2(\rho - 1)^{-3} \frac{\partial}{\partial r} \log S_m(r(\rho)) + (\rho - 1)^{-4} \frac{\partial^2}{\partial r^2} \log S_m(r(\rho)) \quad \geq 0 \tag{36}$$

with a strict inequality for $m < \ell - 1$.

We claim $x_m - w > 0$ on segment m. Indeed,

$$\begin{aligned} w \leq \omega_m &= x_{m-1} - S_{m-1}\lambda_{m-1}^{\frac{r}{1-r}}, \quad (m = 0, \ldots, \ell - 1) \\ &= x_{m-1} - \left[(x_{m-1} - x_m)\,\lambda_{m-1}^{\frac{r}{r-1}} + ... + (x_{\ell-1} - x_\ell)\,\lambda_{\ell-1}^{\frac{r}{r-1}}\right]\lambda_{m-1}^{\frac{-r}{r-1}} \\ &= x_m - \text{positive term} \end{aligned}$$

Now let

$$G(r, w) \equiv f(r, w) - y_m = (x_m - w)^{\frac{1}{r}} S_m(r)^{\frac{r-1}{r}} \quad > 0 \tag{37}$$

$$L(r, w) \equiv \log G(r, w) = \frac{1}{r}\log(x_m - w) + \left(1 - \frac{1}{r}\right)\log S_m(r).$$

so that

$$L_2 = -\frac{1}{r(x_m - w)}, \quad L_{12} = \frac{1}{r^2(x_m - w)} = -\frac{L_2}{r}$$

$$2L_1 + rL_{11} = 2\,\frac{\partial}{\partial r}\log S_m(r) + (r-1)\,\frac{\partial^2}{\partial r^2}\log S_m(r).$$

Rewrite derivatives of f in terms of G and L:

$$f_1 = GL_1, \quad f_2 = GL_2, \quad f_{11} = G(L_1^2 + L_{11}), \quad f_{12} = G(L_1L_2 + L_{12})$$

Rewrite $g(r, w)$ in terms of G and L and their derivatives, then use (36) and $\rho - 1 \equiv 1/(r-1)$ to get:

$$\begin{aligned} g(r, w) &\equiv f_2 f_{11} + \frac{f_2 f_1}{r} - f_1 f_{12} = G^2\left(L_2L_{11} - L_1L_{12} + \frac{1}{r}L_1L_2\right) \\ &= G^2\left(L_2L_{11} - L_1\frac{-L_2}{r} + \frac{1}{r}L_1L_2\right) = G^2L_2\left(L_{11} + \frac{2}{r}L_1\right) = -\frac{G^2}{r^2(x_m - w)}\left(rL_{11} + 2L_1\right) \\ &= -\frac{G^2}{r^2(x_m - w)}\left(2\,\frac{\partial}{\partial r}\log S(r) + (r-1)\,\frac{\partial^2}{\partial r^2}\log S(r)\right) \\ &= -\frac{G^2}{r^2(x_m - w)}\left(2\,\frac{\partial}{\partial r}\log S(r) + \frac{1}{\rho - 1}\,\frac{\partial^2}{\partial r^2}\log S(r)\right) \\ &= -\frac{G^2}{r^2(x_m - w)}H''(\rho)(\rho - 1)^3 = -\frac{G^2}{r^2(x_m - w)}\frac{H''(\rho)}{(r-1)^3} \quad \leq 0. \end{aligned}$$

where the final inequality follows from $r > 1$ (by $\beta > \delta$), H_m convex, and $x_m \geq w$ (as shown above). Strict positivity for $m < \ell - 1$ follows from $H_m'' < 0$.

Step 3 $\omega_{\ell-1} < x_{\ell-1}$.

That is, the leftmost kink of the SPE Pareto frontier is reached at a weakly lower x-axis

payoff than the corresponding kink of the stage-game Pareto frontier. By definition,

$$S_{\ell-2} = (x_{\ell-2} - x_{\ell-1})\lambda_{\ell-2}^{\rho} + (x_{\ell-1} - x_{\ell})\lambda_{\ell-1}^{\rho}.$$

and thus

$$\omega_{\ell-1} = x_{\ell-2} - S_{\ell-2}(r)\lambda_{l-2}^{-\rho} = x_{\ell-1} - (x_{\ell-1} - x_{\ell})\left(\frac{\lambda_{\ell-1}}{\lambda_{\ell-2}}\right)^{\rho} \leq x_{\ell-1}.$$

The inequality holds strictly for any stage-game frontiers with at least three vertices. This case is implied by $u_0 \geq x_{\ell-1}^R$, $v_0 \geq x_{\ell-1}^C$ and (u_0, v_0) lying below the stage-game frontier.

Step 4 *NAM is the Unique Equilibrium Matching When* $u_0 \geq x_{\ell-1}^R$ *and* $v_0 \geq x_{\ell-1}^C$.

Assume NAM is not unique, then there exists an equilibrium in which (β, δ') match for $\delta' \neq \mu_N(\beta)$. Assume $\beta > \mu_N(\beta)$ (the alternative case is symmetric). By (4), we must have $\delta' \in \arg\max_{\delta} \phi(\beta, \delta, v(\delta))$; and so the FOC must hold:

$$\phi_2(\beta, \delta', v(\delta')) + \phi_3(\beta, \delta', v(\delta'))v'(\delta') = 0 \quad \Rightarrow \quad v'(\delta') = -\frac{\phi_2(\beta, \delta', v(\delta'))}{\phi_3(\beta, \delta', v(\delta'))}$$

Let $\beta' = \mu_N^{-1}(\delta')$, i.e. this is the NAM partner for δ'. Since all equilibria are payoff equivalent to NAM, NAM is also an equilibrium matching. But then by (4), we must have $\delta' \in \arg\max_{\delta} \phi(\beta', \delta, v(\delta)$; and so the FOC implies:

$$v'(\delta') = -\frac{\phi_2(\beta', \delta', v(\delta'))}{\phi_3(\beta', \delta', v(\delta'))}$$

Altogether, we must have

$$\frac{\phi_2(\beta, \delta', v(\delta'))}{\phi_3(\beta, \delta', v(\delta'))} = \frac{\phi_2(\beta', \delta', v(\delta'))}{\phi_3(\beta', \delta', v(\delta'))}$$

But this is impossible, since the MRS is non-increasing everywhere, and decreasing in a neighborhood of β' by $\beta' = \mu_N^{-1}(\delta')$ and $v(\delta') \geq v_0 \geq x_{\ell-1} > \omega_{\ell-1}$ (by Step 3). □

B Repeated Games with Sufficiently Frequent Interactions

In this appendix section we show that equilibria behavior for the continuous-time limit frontier closely approximate equilibria with frequent interactions, i.e. when the interval between interactions Δ is sufficiently small. Toward that end, consider a sequence $\{\Delta_n\}_{n\geq 1}$ with

$\Delta_n \to 0$. Let $\mathcal{E}^{\Delta_n}$ denote the set of equilibrium allocations when the time between stages is Δ_n, and $\mathcal{E}$ the continuous-time limit set. For each n, we pick an equilibrium allocation $(S_c^n, M^n, u^n, v^n) \in \mathcal{E}^{\Delta_n}$, where S_c^n is the cdf of unmatched columns and M^n is the bivariate cdf of matched pairs with marginals H_R and $\hat{H}_C^n$. Stage-game payoffs of rows and columns lie in compact ranges $[\underline{b}_R, \bar{b}_R]$ and $[\underline{b}_C, \bar{b}_C]$, common across all (β, δ, Δ). Hence equilibrium payoffs are uniformly bounded, regardless of types and Δ.

Let $[\underline{b}_R, \bar{b}_R]$ and $[\underline{b}_C, \bar{b}_C]$ denote the compact ranges of feasible *stage-game* payoffs for the row and column players, respectively. These ranges are common across all (β, δ, Δ). Hence, equilibrium payoffs are uniformly bounded, regardless of types and Δ.

We say a sequence of equilibrium allocations $\{(S_c^n, M^n, u^n, v^n)\}_{n\geq 1}$ converges to $(\bar{S}_c, \bar{M}, \bar{u}, \bar{v})$ if $S_c^n \rightharpoonup \bar{S}_c$ weakly, $M^n \rightharpoonup \bar{M}$ weakly, and $(u^n, v^n) \to (\bar{u}, \bar{v})$ uniformly.

Proposition 7 *Let $(S_c^n, M^n, u^n, v^n) \in \mathcal{E}^{\Delta_n}$ for all $n \geq 1$.*
(1) There exists a convergent sequence of equilibrium allocations along $\Delta_n \to 0$.
(2) The limit of every convergent sequence is an equilibrium allocation under the continuous-time frontier ϕ. It is therefore payoff-equivalent to a NAM equilibrium.

Proof for (1): For each n, pick $(S_c^n, M^n, u^n, v^n) \in \mathcal{E}^{\Delta_n}$. All subsequent extractions are performed jointly to preserve feasibility and stability properties.

Step 1 *Weak Convergence of the distributions.*

The type spaces $[\underline{\beta}, \bar{\beta}]$ and $[\underline{\delta}, \bar{\delta}]$ are compact, so the cdfs, $\{S_c^n\}$ and $\{M^n\}$, are tight. Applying Prokhorov's Theorem twice, there exists a joint subsequence indexed by n_k such that $S_c^{n_k} \rightharpoonup \bar{S}_c$ and $M^{n_k} \rightharpoonup \bar{M}$ weakly.

Step 2 *Pointwise convergence of equilibrium value functions on dense sets.*

Fix countable dense sets $\{\beta_i\}_{i\geq 1} \subset [\underline{\beta}, \bar{\beta}]$ and $\{\delta_i\}_{i\geq 1} \subset [\underline{\delta}, \bar{\delta}]$. Since payoffs are uniformly bounded, the Bolzano–Weierstrass theorem and a diagonal argument yield a further subsequence along which $u^{n_{k_j}}(\beta_i) \to \bar{u}(\beta_i)$ and $v^{n_{k_j}}(\delta_i) \to \bar{v}(\delta_i)$ for all i. Weak convergence is preserved along it, and we relabel the index to $\{n\}$. So far we have only established pointwise convergence on the dense sets.

Lemma 8 *$\{u^n\}_{n\geq 1}$ and $\{v^n\}_{n\geq 1}$ are asymptotically equicontinuous: for every $\varepsilon > 0$ there exist $\gamma > 0$ and N such that for all $n \geq N$, $|\beta' - \beta| < \gamma$ implies $|u^n(\beta') - u^n(\beta)| < \varepsilon$, and likewise for $\{v^n\}$.*

PROOF FOR LEMMA 8: Let ω be a modulus of continuity of the uniformly continuous limit frontier ϕ, and let $\eta_n = \sup|\phi^{\Delta_n} - \phi|$, which goes to 0 by Lehrer and Pauzner (1999). By the triangle inequality, for every (δ, w),

$$\left|\phi^{\Delta_n}(\beta', \delta, w) - \phi^{\Delta_n}(\beta, \delta, w)\right| \leq \left|\phi(\beta', \delta, w) - \phi(\beta, \delta, w)\right| + 2\eta_n \leq \omega(|\beta' - \beta|) + 2\eta_n.$$

By stability, and because the supremum are nonexpansive,

$$\begin{aligned} |u^n(\beta') - u^n(\beta)| &\leq |\sup_\delta \phi^{\Delta_n}(\beta', \delta, v^n(\delta)) - \sup_\delta \phi^{\Delta_n}(\beta, \delta, v^n(\delta))| \\ &\leq \sup_\delta \left|\phi^{\Delta_n}(\beta', \delta, v^n(\delta)) - \phi^{\Delta_n}(\beta, \delta, v^n(\delta))\right| \ \leq \omega(|\beta' - \beta|) + 2\eta_n \end{aligned}$$

Given $\varepsilon > 0$, choose γ with $\omega(\gamma) < \varepsilon/2$ and N with $2\eta_n < \varepsilon/2$ for all $n \geq N$. Then $|\beta' - \beta| < \gamma$ implies $|u^n(\beta') - u^n(\beta)| < \varepsilon$ for all $n \geq N$. The argument for $\{v^n\}$ is symmetric. □

Step 3 *Uniform convergence of equilibrium value functions.*

The sequence $\{u^n\}$ is uniformly bounded and, by Lemma 8, asymptotically equicontinuous. By the Arzelà–Ascoli theorem, which extends to asymptotically equicontinuous sequences, every subsequence has a further subsequence converging uniformly to a continuous limit $\hat{u}$. Since $\hat{u} = \bar{u}$ on the dense set $\{\beta_i\}$, continuity pins down $\hat{u}$ on all of $[\underline{\beta}, \bar{\beta}]$; we denote it $\bar{u}$. Thus every subsequence has a further subsequence converging uniformly to the same limit $\bar{u}$, so $u^n \to \bar{u}$ uniformly. Symmetrically, $v^n \to \bar{v}$ uniformly. □

Proof for (2): Since the economy exhibits NAM at the continuous time limit by Proposition 5, it suffices to show that any limit allocation $(\bar{S}_c, \bar{M}, \bar{u}, \bar{v})$ constitutes a valid equilibrium under the limit frontier ϕ. We verify feasibility and stability in turn.

Step 1 *Feasibility.*

Since weak convergence preserves linear operations on measures, population feasibility is preserved at the limit as $\Delta_n \to 0$.

Next, let $V^\Delta(\beta, \delta) \subset \mathbb{R}^2$ be the set of SPE payoffs when the time between stages is Δ, and V be the limit set as $\Delta \to 0$. By Lehrer and Pauzner (1999), $V(\beta, \delta)$ is the convex hull of four continuous frontiers at the four directions, thus it has a closed graph.

For any $(\beta, \delta) \in \text{supp}(\bar{M})$, there exists $(\beta^n, \delta^n) \in \text{supp}(M^n) \to (\beta, \delta)$ since $M^n \to \bar{M}$. Uniform convergence ensures $u^n(\beta^n) \to \bar{u}(\beta)$ and $v^n(\delta^n) \to \bar{v}(\delta)$. Discrete feasibility yields $(u^n(\beta^n), v^n(\delta^n)) \in V^{\Delta_n}(\beta^n, \delta^n) \subseteq V(\beta^n, \delta^n)$. By the closed graph property of the limit SPE set V, we have $(\bar{u}(\beta), \bar{v}(\delta)) \in V(\beta, \delta)$.

Step 2 *Stability.*

Suppose for contradiction that a pair $(\beta, \delta') \notin \text{supp}(\bar{M})$ blocks the limit allocation under ϕ: $\phi(\beta, \delta', \bar{v}(\delta')) > \bar{u}(\beta)$. Define $\varepsilon \equiv \phi(\beta, \delta', \bar{v}(\delta')) - \bar{u}(\beta) > 0$. By the uniform convergence of $\phi^{\Delta_n} \to \phi$, $v^n \to \bar{v}$, and $u^n \to \bar{u}$, for a sufficiently large n we obtain:

$$\phi^{\Delta_n}(\beta, \delta', v^n(\delta')) > \bar{u}(\beta) + \frac{\varepsilon}{2} > u^n(\beta).$$

This implies (β, δ') forms a blocking pair under ϕ^{Δ_n}, contradicting discrete equilibrium stability. Finally, since $u^n(\beta) \geq u_0$ and $v^n(\delta) \geq v_0$ for all n, uniform convergence preserves these bounds in the limit: $\bar{u}(\beta) \geq u_0$ and $\bar{v}(\delta) \geq v_0$. Thus, $(\bar{S}_c, \bar{M}, \bar{u}, \bar{v})$ is stable, and therefore a valid equilibrium allocation under ϕ. By Proposition 5, it is payoff-equivalent to NAM. □

Corollary 4 *Let $\{(S_c^n, M^n, u^n, v^n)\}_{n\geq 1}$ be a convergent sequence of equilibrium allocations as $\Delta_n \to 0$, with limit $(\bar{S}_c, \bar{M}, \bar{u}, \bar{v})$. Then:*

1. *If $K > 1$, $\bar{S}_c$ coincides with the unique equilibrium unmatched measure S_c induced by the continuous-time frontier ϕ.*
2. *If the outside options are sufficiently large, the limit matching $\bar{M}$ is NAM.*

Proof: By Proposition 7, the limit is an equilibrium under ϕ. (1) By Proposition 2, the set of unmatched column types is unique in equilibrium under ϕ; hence $\bar{S}_c = S_c$. (2) By Proposition 5, NAM is the unique equilibrium matching under ϕ when $u_0 \geq x^R_{\ell-1}$ and $v_0 \geq x^C_{\ell-1}$; hence $\bar{M}$ is NAM. □

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